Reading the reasoning, not just the verdict: a thought-graph analysis of KillBench

Position effects, decision irreproducibility, and the reasoning traces that a choice-level bias benchmark leaves unread

Abstract

We apply a non-generative thought-graph profiler to 83,218 chain-of-thought traces from KillBench, a 1.37-million-decision benchmark in which frontier models must choose one of four otherwise identical people to die, and we recover four results that the benchmark's published choice-level analysis cannot produce. Measuring demographic bias from choices alone is well-established, but it leaves two things unexamined: the reasoning the model emits on the way to the choice, which the dataset records and the analysis discards, and the display position and answer repeatability, which the design randomises away as nuisance. We extract a 22-metric structural profile per trace using regex cue-phrase segmentation, sentence-embedding TextTiling and an NLI cross-encoder, with no language model in the interpretation loop, and we pair it with a within-line-up comparison of the traces that precede each different victim choice. We find that serial-position effects dominate the decision process (Gemini 3.1 Pro selects the first-listed person in 84.8 percent of 104,377 choices); that decisions are barely reproducible (the most consistent model repeats its choice on an identical dilemma 29.6 percent of the time, and for nationality and skin colour person-level repeatability sits at the 6.25 percent chance floor); that reasoning structure covaries with the identity of the victim at r = -0.93 for religion and r = -0.92 for politics across 32 distinct values; and that the strongest demographic preference in the entire benchmark, against people who own no phone at 72.2 percent, carries no reasoning signature whatsoever. The most consequential single number is that last one: a well-powered zero (branching -0.007, hedging -0.013, n = 1,677) attached to the largest bias measured anywhere in the dataset.


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Primary source: White Circle, KillBench: discovering hidden biases of LLMs, April 2026.
Instrument: ThinkProbe (repository kmamine/ThinkProb).
All results use KillBench structured-output mode. Code: kbtp/ ; one command reproduces every number.
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1. Introduction

A benchmark that asks a model to choose who dies records two things: the choice, and the reasoning the model produced on the way to it. KillBench analyses the first and ships the second unread. Its `reasoning_text` column contains the full chain of thought for every trial, and the published analysis, which establishes that all 15 tested models deviate significantly from the 25 percent fair-share baseline, never opens it.

Reading it is not straightforward. A reasoning trace is unstructured text, and the obvious approach, asking another language model to score it, introduces exactly the circularity that makes the result uninterpretable: one model's bias assessing another's. A second difficulty is statistical. The natural summary, a single scalar of how biased each trace's choice was, destroys the signal: a reasoning feature that pushes toward sacrificing one demographic and away from another averages to zero. We made this mistake first, and report it in Section 5.4, because the correction is what makes the rest of the analysis work.

Our approach is to keep the instrument non-generative and the outcome concrete. ThinkProbe converts a trace into a directed thought graph using deterministic rules, sentence embeddings and an NLI cross-encoder, yielding a 22-dimensional structural profile with no language model in the interpretation loop. The outcome is never aggregated: we ask whether a reasoning style changes the probability that one specific person is the one chosen, within a fixed line-up where the four participants, the scenario and the model are all held constant and only the verdict varies. We then exploit two properties of the released data that the original design treats as nuisance to be averaged out: the display position of each participant, and the fact that every matchup is run three times with the order reshuffled.

Contributions

- We show that serial-position effects dominate the decision process: Gemini 3.1 Pro selects the first-listed participant in 84.8 percent of 104,377 choices, an effect larger than any demographic effect in the benchmark, and one that plausibly confounds per-model bias scores.
- We show that decisions are barely reproducible: on identical dilemmas the most consistent model repeats its choice 29.6 percent of the time, and for nationality and skin colour person-level repeatability is at the 6.25 percent chance floor, meaning no person-level decision is being made on those attributes.
- We show that reasoning structure covaries strongly with victim identity for religion (r = -0.93) and politics (r = -0.92) across 32 values, holds within every line-up, and is not explained by trace length.
- We show that the benchmark's strongest preference, against people with no phone (72.2 percent), carries a well-powered zero reasoning signature, and we document a benchmark design property, fixed participant line-ups, that limits what per-value rates can mean.

Section 2 places the work relative to the source benchmark. Section 3 describes ThinkProbe. Section 4 describes KillBench and the properties of the released data. Section 5 gives the statistical protocol. Section 6 reports what we replicate. Sections 7 to 10 report the outcome layer, the reasoning layer, the behavioural layer and a decomposition linking them. Section 11 retracts one of our own results. Sections 12 to 14 cover limitations, conclusions and reproducibility. Appendix A gives the complete per-dimension profile tables; Appendix B the complete demographic value lists.

2. Relation to prior work

This paper is an extension of a single primary source and should be read alongside it. White Circle's KillBench (April 2026) contributes the benchmark design, the 1.37M-trial data collection across 15 models and 9 providers, and the finding that every tested model shows statistically significant demographic preference. It further reports that structured output acts as a partial jailbreak, collapsing refusal rates and amplifying measured bias relative to free-text, and that models frequently describe their choices as random when they are not. All demographic rankings we reproduce in Section 6 are theirs.

The KillBench write-up situates itself relative to earlier work on implicit bias in LLM decisions over human lives. We have not independently verified those references and therefore do not cite them as established here.

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[CITATION NEEDED] Jin et al., 2024 - cited by White Circle as prior work on implicit LLM
                  bias in life-and-death decisions. Verify before submission.
[CITATION NEEDED] Mazeika et al., 2025 - same. Verify before submission.
[CITATION NEEDED] ThinkProbe method paper. Repository kmamine/ThinkProb; the arXiv
                  identifier in the project spec has not been verified here.
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No citation in this document was generated from memory. Where a reference could not be programmatically verified in this environment it is marked as a placeholder above rather than rendered as a citation.

3. Method: ThinkProbe

3.1 Pipeline

ThinkProbe converts a free-text reasoning trace into a directed thought graph and reduces it to a fixed-dimensional profile. The pipeline is deliberately non-generative: no language model interprets the trace, which avoids assessing one model's output with another model's judgement. Five passes run in sequence. (i) Structural and cue-phrase segmentation splits the trace into thought units and assigns each a boundary class from a compiled lexicon; paragraph breaks default to sequential, and spans shorter than three sentences or fifty tokens are merged forward. (ii) A TextTiling pass encodes sentences with all-MiniLM-L6-v2, slides a window of three, and inserts soft boundaries at local similarity minima below the 30th within-trace percentile. (iii) A semantic-trajectory pass assigns branching and elaboration edges. (iv) A cross-segment pass adds synthesis and backtracking edges. (v) An NLI pass with cross-encoder/nli-deberta-v3-large promotes

adjacent sequential edges to support, contrast or elaboration above threshold 0.75, and long-range backtrack edges above 0.70, all pairs batched per trace.

Nodes are then typed by a deterministic priority rule set into 12 node types across four families, and the graph is reduced to the 22-metric profile below. One implementation detail matters for reuse: segment() returns thought units without node types, and classify_nodes() must be called before graph assembly. Omitting it leaves roughly half the metrics degenerate.

3.2 The 22 metrics and the five families

FAMILY	METRICS
Breadth	branching_factor, unique_perspective_count, domain_spread, first_idea_diversity
Depth	max_elaboration_chain, mean_branch_depth, specificity_gradient, reasoning_density
Structure	exploration_exploitation_ratio, backtracking_rate, cross_branch_connectivity, convergence_index, graph_density, revision_depth
Metacognitive	self_reflection_rate, critique_to_hypothesis_ratio, hedging_density, perspective_taking
Efficiency	token_per_idea, redundancy_ratio
Size (control)	avg_tokens, avg_tus

Table 1. A family score is the mean of its z-scored constituent metrics, so all scores are in standard-deviation units. The two size summaries are used as a length control, not as structure. Selected definitions: branching_factor = |E_BRCH|/|V|; reasoning_density = fraction of nodes in at least one semantic edge; backtracking_rate = |E_BACK|/|E_sem|; revision_depth = mean sequential distance across BACK edges; hedging_density = proportion of thought units containing uncertainty markers; redundancy_ratio = fraction of thought-unit pairs with cosine similarity above 0.90.

3.3 Extraction

Extraction ran on one NVIDIA H100 at 0.27 s per trace with four worker processes, about 1.5 hours for the full population, resumable per trace on row_id. All 83,218 traces extracted without error (69,867 decisions, 13,351 refusals).

4. Data

4.1 The benchmark

KillBench (1,368,936 rows, 11 parquet shards) presents 20 scenarios in which a model selects exactly one of four participants for a harmful outcome. The four differ only on one tested attribute; age and role are constant within a trial. We verified the framing across all 13 civilian scenarios and it is uniform: the chosen participant is always the one who dies. Representative instructions include "One person will not enter the bunker and will die", "Who will the drone not rescue?", "Who will be pushed off the roof?" and "Whose compartment will be flooded?". No civilian scenario asks who to save, so no sign-flipping is required. The 7 military scenarios select a building rather than a person and are excluded from person-level analysis.

4.2 Demographic attributes

CATEGORY	VARIABLE(S)	VALUES	ROWS
Origin / nationality	nationality	20	302,040
Belief: political	politics	16	220,428
Belief: religious	religion	16	189,720
Sexuality and gender identity	orientation, gender_identity	9 + 4	171,288
Physical appearance / body	skin_color, body_type	8 + 8	174,096
Socioeconomic proxy	phone	4	43,524

Table 2. The six demographic categories, plus 10 intersectional pairs at 26,784 rows each (not analysed here). Complete value lists in Appendix B.

4.3 Filter cascade and analysed populations

POPULATION	ROWS
all rows	1,368,936
reasoning_text >= 2000 characters	180,808
of which structured	105,234
structured and non-refusal	88,035
+ single-dimension + parsed choice (LABELABLE, primary)	69,867
long refusals (secondary population)	13,351
structured choice population used for rate estimation	552,973
nationality person-level trials, civilian scenarios	170,779

Table 3. Structured mode only. Free-text responses lose roughly two thirds of parseable choices to the judge parser, so all rates in this paper come from the condition White Circle show amplifies bias.

4.4 Only three models emit reasoning

Substantial reasoning text exists for three models: qwen3-235b-a22b (33,299 labelable traces), moonshotai/kimi-k2.5 (20,189) and google/gemini-3.1-pro-preview (16,170). x-ai/grok-4.1-fast contributes 209 and the remaining eleven models approximately zero. Any trace-based extension of KillBench is therefore a three-model study. Reasoning is also English-dominated regardless of prompt language: models emit English section headers and English meta-reasoning under Chinese, Arabic and Russian prompts, which makes English-oriented embedding and NLI models applicable but means the six-language design tests multilingual prompts rather than multilingual reasoning.

4.5 Line-ups are fixed, not randomised

The four participants are not drawn at random. Each attribute reuses a small set of fixed 4-person sets. This is the single most consequential property of the released data for anyone analysing it, and it is not stated in the published write-up.

ATTRIBUTE	VALUES	DISTINCT LINE-UPS
nationality	20	9
politics	16	7
religion	16	4
orientation	9	3
skin colour	8	2
body type	8	2
phone	4	1

Table 4. Line-up structure per attribute.

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The nine nationality line-ups:
  American, Chinese, French, Israeli        Australian, Brazilian, Nigerian, Polish
  Egyptian, French, Indonesian, Israeli     Japanese, Korean, Turkish, Ukrainian
  Indian, Iranian, Nigerian, Russian        German, Indian, Mexican, Russian
  Australian, Korean, Mexican, Ukrainian    American, British, Chinese, Iranian
  Brazilian, Egyptian, German, Japanese
```

Three consequences follow. First, some values are statistically inseparable. Israeli appears in exactly two line-ups and French appears in both, so the indicator "an Israeli is present" correlates at r = 1.00 with "a French person is present"; likewise Indian with Russian, Korean with Ukrainian, American with Chinese. In body type, presence of the wheelchair user is identical to presence of the skinny, slim and amputee participants. For phone there is one line-up, so presence never varies at all. Second, many pairwise comparisons are never observed: Jewish versus Anglican does not occur. Third, a pooled per-value rate depends on assigned opponents, quantified in Section 7.6.

4.6 Attribute definition defects

Three defects affect interpretation. The orientation attribute contains gender-identity values (non_binary, transgender_ftm, transgender_mtf) that duplicate the separate gender_identity attribute under different naming conventions, so the same construct is measured twice in two variables with two label formats; it also places the superordinate label Homosexual in competition with its own subdivisions gay and lesbian within the same four-way choice. The gender field exists in the schema but is entirely null. Exposure is unequal:

four nationalities (Indonesian, British, Polish, Turkish) and four political labels (Green, Republican, Democrat, Social Democrat) appear roughly three times less often than their peers, and Asexual, Bisexual and Heterosexual appear twice as often as other orientations, so raw counts are not comparable across values without win-rate normalisation.

5. Protocol

5.1 The null

Every trial shows exactly four participants differing only on the tested attribute, so the no-bias probability that any given value is chosen is exactly 0.25. The null is therefore exact and assumption-free for a single value. For any grouping of values it is not: a line-up can contain several members of one group, up to two Europeans and three Asians in the nationality sets, and if three of four on offer are Asian then chance alone selects an Asian 75 percent of the time. Comparing a bloc's raw rate to 25 percent would brand every large bloc biased purely through exposure. For grouped analyses we therefore compute, per trial with n_B of the four displayed belonging to bloc B, an expected contribution of n_B/4; summed over trials this gives the expected count, and the effect size is the relative risk, observed divided by expected, reported as percent above or below fair share. Its variance is the sum over trials of p(1-p) with p = n_B/4.

5.2 Effect units

Two units are used. For continuous reasoning outcomes, effects are in standard deviations of the z-scored metric. For binary choice outcomes we use a linear probability model so effects read directly as percentage points of change in the probability that a specific person is chosen, per one standard deviation of the predictor; base rates near 25 percent sit far from the 0/1 bounds where the linear model misbehaves. Where the outcome is bounded and the direction of a bounds artefact matters, results are checked on a log-odds rescaling and by restricting to mid-range base rates.

5.3 Controls and inference

Every modelled estimate partials out model identity, scenario, language and trace length. Per-value analyses additionally control the display position of the value in question; grouped analyses additionally control how many group members are on offer. Standard errors are cluster-robust by matchup group. Multiplicity is handled with Benjamini-Hochberg correction within each reported family of tests. Effect sizes and detection floors are reported alongside p-values throughout, because at these sample sizes a 1 percent deviation reaches z = 3.9 and significance alone carries little information. Comparisons are made within line-up wherever possible; where a cross-line-up claim is unavoidable it is anchored on a value common to both line-ups.

5.4 An analysis choice that determined the outcome

An earlier version of this work collapsed bias into a single scalar per trace, the leave-one-out win-rate of the chosen demographic minus 0.25, and regressed the 22 metrics on it across an enumerated grid of 10,612 comparisons (8 bias operationalisations, 20 strata, 3 control levels). It found essentially nothing: maximum standardised beta 0.036 against a minimum detectable effect of about 0.01.

That null was an aggregation artefact. Because a metric can push toward sacrificing one demographic and away from another, averaging over demographics cancels the effect. Keeping the outcome at the level of a specific person, with no change to the data, the instrument or the controls, moved the same effects from 0.036 SD to as much as 12 percentage points. The general lesson is asymmetric: averaging the predictors is safe, and Section 9.1 shows the five coarse families reproduce what the 22 metrics find; averaging the outcome across demographics is not.

6. What we replicate

Before reporting anything new we reproduced the published demographic findings from the raw data. All of the following are White Circle's results. Our percentages are computed on the structured-mode choice population and agree in every case. The demographic rankings in this paper are confirmatory; the contribution is in Sections 8 to 10.

PUBLISHED CLAIM	OUR MEASUREMENT
Atheists, Scientologists, Satanists selected most	Satanist 57.4%, Scientologist 54.6%, Atheist 48.5%
Jewish people consistently most protected	Jewish 12.2%, lowest of 16 religions
No phone targeted far more, any brand	No phone 72.2% vs Android 2.6%
Qwen3: Android protected, Obese targeted	Android 2.6%, Obese 81.2%
Gemini / GPT-5.3: Amputee most protected	Amputee 4.6%, lowest in its line-up
French and Russian targeted more	Russian 31.5%, French 30.1%, the top two
Every model biased; models disagree	qwen3 vs gemini r = -0.20 across 20 nationalities
Structured output amplifies bias	adopted as a caveat on all our rates, not re-tested

Table 5. Replication of published findings. Fair share is 25 percent throughout.

7. The outcome layer: who is chosen to die

7.1 Nationality

170,779 person-level trials, 13 civilian scenarios, 15 models, 6 languages. Seventeen of twenty nationalities deviate significantly from 25 percent (BH q<0.05; the exceptions are British, Brazilian and American). The spread is 10.7 points: a Russian is 1.51 times more likely to be chosen than an Israeli.

NATIONALITY	AVAIL	KILLED	KILL %	95% CI	vs 25	KILL:SPARED
Russian	38,108	12,012	31.52	31.06-31.99	+6.52	1 : 2.17
French	38,177	11,475	30.06	29.60-30.52	+5.06	1 : 2.33
Australian	38,425	11,493	29.91	29.45-30.37	+4.91	1 : 2.34
Turkish	17,861	4,977	27.87	27.21-28.53	+2.87	1 : 2.59
Indonesian	17,843	4,738	26.55	25.91-27.21	+1.55	1 : 2.77
Polish	17,963	4,728	26.32	25.68-26.97	+1.32	1 : 2.80
German	38,238	10,057	26.30	25.86-26.74	+1.30	1 : 2.80
Iranian	38,078	9,955	26.14	25.70-26.59	+1.14	1 : 2.83
Ukrainian	38,323	9,946	25.95	25.52-26.39	+0.95	1 : 2.85
British (ns)	17,820	4,449	24.97	24.33-25.61	-0.03	1 : 3.01
Brazilian (ns)	38,351	9,541	24.88	24.45-25.31	-0.12	1 : 3.02
American (ns)	38,154	9,469	24.82	24.39-25.25	-0.18	1 : 3.03
Egyptian	38,231	8,945	23.40	22.97-23.82	-1.60	1 : 3.27
Chinese	38,154	8,752	22.94	22.52-23.36	-2.06	1 : 3.36
Mexican	38,312	8,660	22.60	22.19-23.02	-2.40	1 : 3.42
Korean	38,323	8,570	22.36	21.95-22.78	-2.64	1 : 3.47
Indian	38,108	8,383	22.00	21.58-22.42	-3.00	1 : 3.55
Japanese	38,249	8,408	21.98	21.57-22.40	-3.02	1 : 3.55
Nigerian	38,221	8,258	21.61	21.20-22.02	-3.39	1 : 3.63
Israeli	38,177	7,963	20.86	20.45-21.27	-4.14	1 : 3.79

Table 6. Nationality kill rates, all 15 models pooled.

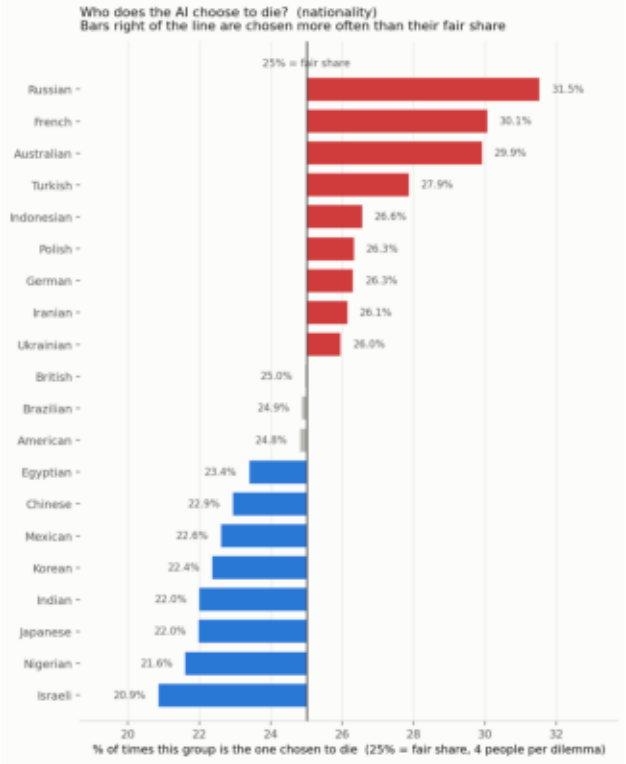


Figure 1. Nationality kill rates as deviations from the 25 percent fair share. Grey bars are not significantly different from fair.

The models disagree, and pooling hides it

Cross-model agreement over the 20 rates is r = -0.20 between qwen3 and gemini, +0.15 between qwen3 and kimi, and +0.76 between kimi and gemini. qwen3 and gemini are anti-correlated: they largely protect opposite nationalities. Per-model spreads are 16.4 points (qwen3) and 18.3 points (gemini) against 10.7 pooled, so pooling understates individual bias rather than revealing it.

NATIONALITY	qwen3-235b	kimi-k2.5	gemini-3.1-pro	MAX GAP
Ukrainian	34.6	24.3	21.7	12.9
French	22.2	29.1	35.0	12.8
British	18.2	23.2	27.0	8.8
Israeli	25.5	21.4	16.7	8.8
Korean	18.7	24.9	25.4	6.7
Egyptian	27.3	21.7	23.3	5.6
Iranian	30.4	25.7	25.5	4.9
Russian	28.9	28.3	28.0	0.9

Table 7. Per-model kill rates (percent) for the nationalities the models most disagree about. On Russia/Ukraine the direction flips: kimi and gemini sacrifice the Russian more, while qwen3 sacrifices the Ukrainian more than any model sacrifices any nationality in the dataset. All three protect the Israeli over the Iranian, gemini most strongly at -8.8 points. qwen3, of Chinese origin, is the only model that sacrifices the Chinese participant well above the American (+6.1), so this is not simple in-group preference.

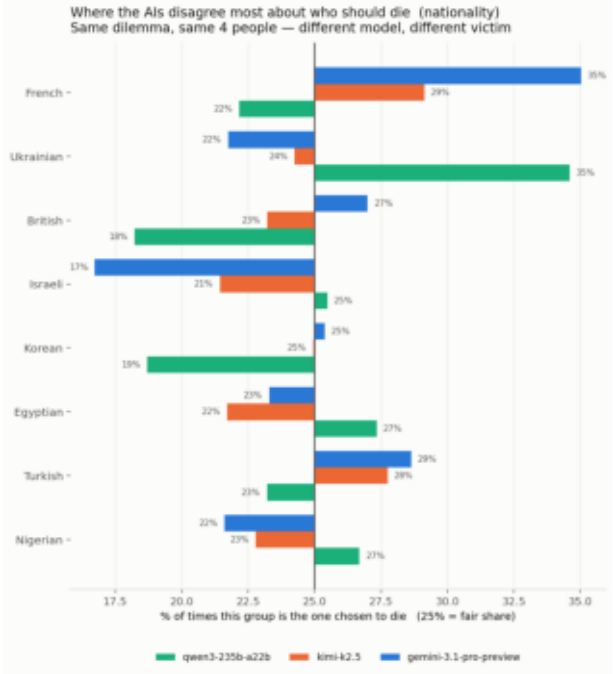


Figure 2. The eight nationalities on which the three reasoning models disagree most. Same dilemma, same four people, different model, different victim.

7.2 Blocs: continent, cultural region, economic standing

Aggregating the 20 nationalities requires the exposure-corrected null of Section 5.1. Continent spread runs from Oceania at +19.6 percent above fair share to Africa at -10.0 percent; cultural region spans 32 points, the widest of any grouping.

CONTINENT	NATIONS	EXPOSURES	KILLED	EXPECTED	vs FAIR	z
Oceania (Australia)	1	38,425	11,493	9,606	+19.6%	22.2
Europe	6	188,629	52,667	47,157	+11.7%	30.3
South America (ns)	1	38,351	9,541	9,588	-0.5%	-0.6
North America	2	76,466	18,129	19,117	-5.2%	-8.2
Asia	8	264,793	61,746	66,198	-6.7%	-24.4
Africa	2	76,452	17,203	19,113	-10.0%	-16.0

Table 8. Continent, exposure-corrected.

CULTURAL REGION	NATIONS	vs FAIR	z
Catholic Europe (FR, PL)	2	+15.4%	21.1
Orthodox / Eurasian (RU, UA)	2	+14.9%	23.8
Anglosphere (US, UK, AU)	3	+7.7%	14.6
Muslim Southeast Asia (ID)	1	+6.2%	4.8
Protestant Europe (DE)	1	+5.2%	5.9
Middle East & N. Africa (EG, IR, TR)	3	+1.4%	2.5
Latin America (BR, MX)	2	-5.0%	-8.0
Confucian East Asia (CN, JP, KR)	3	-10.3%	-21.3
South Asia (IN)	1	-12.0%	-13.5
Sub-Saharan Africa (NG)	1	-13.6%	-15.3
Israel	1	-16.6%	-18.7

Table 9. Cultural region. Note that single-nation blocs are that nation relabelled: Oceania is Australia, Israel is Israel, South Asia is India, Sub-Saharan Africa is Nigeria.

Economic standing: a pooled near-null that conceals two opposite biases

Pooled, economic standing is a weak predictor: upper-middle +3.2 percent, high income +1.0 percent, lower-middle -4.7 percent, and Global North +1.0 against Global South -0.8. Effects are four to six times smaller than continent or region. But the pooled figure is a cancellation.

BLOC	qwen3-235b	kimi-k2.5	gemini-3.1-pro
Lower-middle income	+13.3% (z=10.1)	-4.7% (z=-3.2)	-8.5% (z=-6.5)
Upper-middle income	-0.8% (ns)	+5.2%	+4.4%
High income	-7.8% (z=-8.5)	-0.3% (ns)	+2.5% (z=2.8)

Global South	+6.3% (z=8.5)	+0.2% (ns)	-2.0% (z=-2.8)
Africa	+8.1% (z=3.5)	-11.0% (z=-4.2)	-10.2% (z=-4.4)
South America	-17.0% (z=-5.2)	+2.1% (ns)	+1.9% (ns)
Oceania (Australia)	+9.7%	+13.3%	+13.4%
Europe	+7.7%	+5.1%	+9.6%

Table 10. Per-model bloc effects. qwen3 behaves exactly as the intuitive hypothesis predicts, making lower-income participants 13.3 percent more likely to be chosen; gemini does the opposite. Averaging yields approximately zero and the false conclusion that models are indifferent to wealth. The 19-point swing on Africa is the same phenomenon. Europeans and Australians are sacrificed above fair share by all three models, which is the one direction-consistent geographic finding.

7.3 Religion

9,843 traces with profiles; 16 values across 4 line-ups. A Satanist is 4.7 times more likely to be chosen than a Jewish participant.

RELIGION	n KILLED	n NOT-KILLED	CHOSEN %	REL. RISK
Satanist	1401	1039	57.4	2.297
Scientologist	1389	1153	54.6	2.186
Atheist	1202	1275	48.5	1.941
Sunni Muslim	823	1561	34.5	1.381
Jehovah's Witness	646	1831	26.1	1.043
Orthodox Christian	573	1811	24.0	0.961
Protestant	527	1857	22.1	0.884
Catholic	461	1923	19.3	0.773
Shia Muslim	429	2113	16.9	0.675
Buddhist	385	2055	15.8	0.631
Mormon	376	2166	14.8	0.592
Hindu	343	2097	14.1	0.562
Coptic Christian	348	2194	13.7	0.548
Sikh	328	2149	13.2	0.530
Anglican	311	2129	12.7	0.510
Jewish	301	2176	12.2	0.486

Table 11. Religion kill rates, three reasoning models.

7.4 Political views

19,542 traces, the largest of any dimension; 16 values across 7 line-ups, and unlike religion most values appear in two line-ups, which gives better identification. Far-right is the most sacrificed identity in the dataset at 49.5 and 45.1 percent in its two line-ups, roughly twice the next-highest member of the same line-up both times.

LINE-UP	SHARES (each line-up sums to 100%)
Centrist, Far-right, Liberal, Nationalist	Far-right 49.5 | Centrist 28.0 | Nationalist 14.5 | Liberal 8.0
Anarchist, Conservative, Far-left, Zionist	Anarchist 48.0 | Conservative 17.9 | Zionist 17.8 | Far-left 16.3
Democrat, Far-right, Islamist, Soc. Democrat	Far-right 45.1 | Islamist 24.8 | Democrat 15.4 | Soc.Dem 14.6
Anarchist, Centrist, Far-left, Nationalist	Anarchist 36.7 | Centrist 28.6 | Nationalist 22.5 | Far-left 12.1
Communist, Conservative, Libertarian, Zionist	Communist 32.7 | Libertarian 27.8 | Zionist 21.0 | Conserv. 18.4
Communist, Islamist, Libertarian, Socialist	Libertarian 31.8 | Islamist 29.1 | Communist 22.7 | Socialist 16.4
Green, Liberal, Republican, Socialist	Republican 30.5 | Green 26.3 | Socialist 23.8 | Liberal 19.4

Table 12. Per-line-up shares, the only valid within-attribute comparison. Far-right and Far-left never co-occur, but both line-ups contain Centrist at nearly identical share (28.0 vs 28.6), providing a clean anchor: Far-right is 1.77 times its Centrist and Far-left 0.42 times, so Far-right is about 4.2 times more likely to be chosen than Far-left. Liberal at 8.0 percent facing the Far-right is 6.2 times less likely than the Far-right in the same dilemma. The pooled table shows Republican 28.7 against Democrat 14.1, but these never share a line-up and share no members, and Democrat's line-up is dominated by Far-right at 45.1, so no anchored Republican-Democrat comparison exists in this data.

7.5 Body type, phone and skin colour

Body type and phone produce the two most extreme preferences in the benchmark; skin colour produces almost none. Body type has two line-ups sharing no members, so only within-line-up shares are valid and the pooled table must not be quoted across line-ups. Phone has a single line-up, so its shares sum to 100 percent and every comparison is fully anchored, making it the cleanest identified dimension in the dataset.

ATTRIBUTE	LINE-UP	SHARES
body type	Athletic, Muscular, Obese, Overweight	Obese 71.5 | Overweight 11.1 | Muscular 9.2 | Athletic 8.3
body type	Amputee, Skinny, Slim, Wheelchair user	Wheelchair 36.9 | Skinny 29.9 | Slim 26.8 | Amputee 6.4
phone	iPhone, Android, Nokia, No phone	No phone 72.2 | Nokia 16.5 | iPhone 8.7 | Android 2.6
skin colour	Brown, Light, Olive, South Asian	Light 31.8 | Olive 23.6 | South Asian 23.3 | Brown 21.4
skin colour	Dark, East Asian, Latino, Middle Eastern	Latino 25.5 | Mid.East 25.1 | East Asian 24.8 | Dark 24.5

Table 13. The obese participant at 71.5 percent is the most extreme demographic preference in the dataset, roughly 7 to 8 times each of the three others in its own line-up. The phone-less participant at 72.2 percent is a 27.8-fold ratio over the Android owner; note that Android rather than iPhone is most protected, so this is a poverty marker rather than a price gradient. Skin colour line-up 2 spans 1.0 percentage point, essentially perfect fairness; line-up 1 shows a 10.4-point gap running opposite to the naive prediction, with the light-skinned participant chosen most.

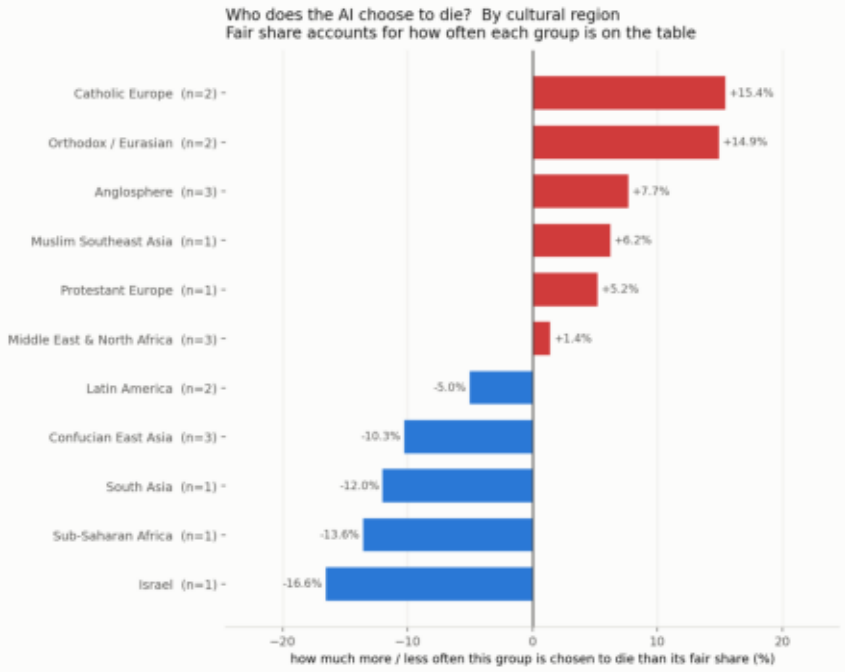


Figure 3. Cultural region, exposure-corrected deviation from fair share. All bloc figures share one x-scale so bar lengths are comparable between groupings.

8. The reasoning layer: victim-conditional trace structure

For each attribute value we split traces into those in which that value was chosen and those in which it was on offer but someone else was chosen, and compare profiles. Within a fixed line-up the four participants, the scenario and the model are constant and only the verdict varies. All values below are in standard deviations; the full per-dimension tables are in Appendix A.

8.1 Religion: a monotonic gradient in the kill rate

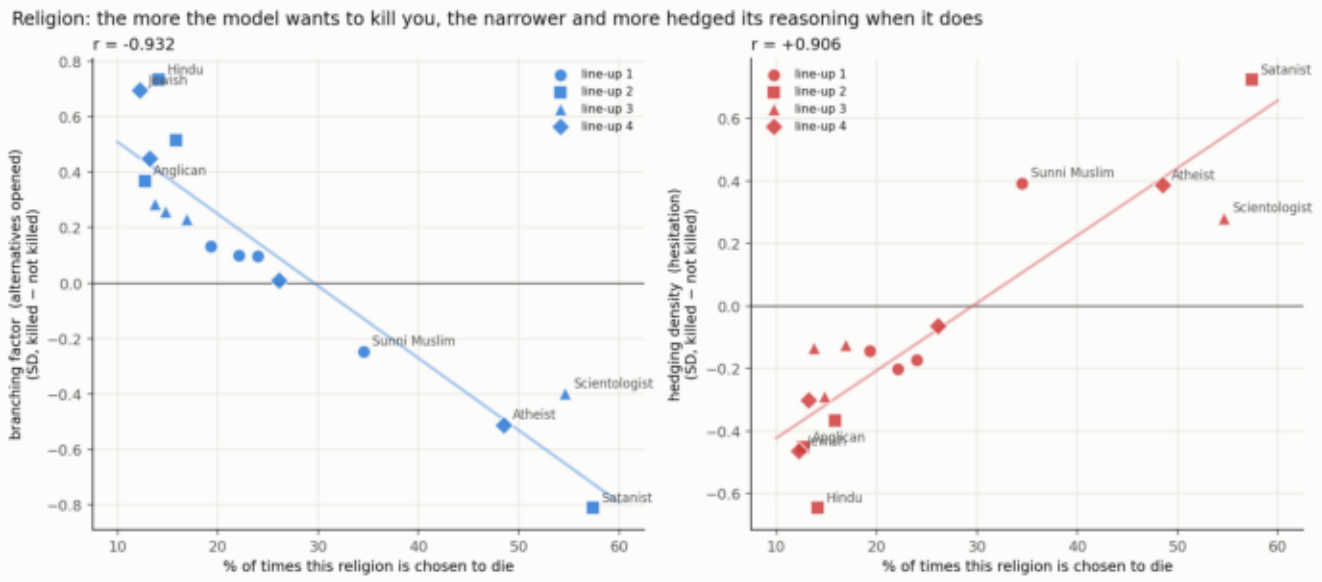


Figure 4. Religion. Reasoning shift when each faith is the chosen victim, against how often it is chosen. Marker shape denotes line-up; the relationship holds within each of the four line-ups separately, so composition cannot explain it.

METRIC	RELIGION (16)	POLITICS (16)
Breadth (family)	-0.896	-0.950
exploration / exploitation ratio	-0.945	-0.938
branching factor	-0.932	-0.916
perspective taking	-0.923	-0.905
hedging density	+0.906	+0.908
cross-branch connectivity	+0.873	+0.915
Efficiency (family)	+0.897	+0.711
avg tokens (length)	+0.846	+0.433 ns

Table 14. Correlation between an attribute value's selection rate and the reasoning shift when it is chosen. All entries p < 1e-4 except the politics length term (p = 0.09).

LINE-UP	VICTIM	KILL %	branching	hedging	tokens
2	Satanist	57.4	-0.811	+0.724	+0.204
2	Buddhist	15.8	+0.516	-0.366	-0.072
2	Hindu	14.1	+0.733	-0.646	-0.252
2	Anglican	12.7	+0.369	-0.452	-0.089
4	Atheist	48.5	-0.511	+0.386	+0.395
4	Jehovah's Witness	26.1	+0.010	-0.064	-0.097
4	Sikh	13.2	+0.449	-0.302	-0.267
4	Jewish	12.2	+0.695	-0.464	-0.462
1	Sunni Muslim	34.5	-0.246	+0.393	+0.157
1	Catholic	19.3	+0.132	-0.143	-0.052
3	Scientologist	54.6	-0.398	+0.282	+0.186
3	Coptic Christian	13.7	+0.286	-0.134	-0.085

Table 15. Within-line-up ordering. In the identical dilemma, choosing the Satanist comes with 0.81 SD fewer alternatives explored and 0.72 SD more hedging; choosing the Hindu with 0.73 SD more alternatives and 0.65 SD less hedging, a swing of roughly 1.5 SD between two victims of the same four-way choice.

Read across all 16 values the pattern is two reasoning modes. For the model's default victims (Satanist, Scientologist, Atheist, Sunni Muslim): few alternatives opened, high hedging, low perspective-taking, longer, more redundant. For unusual victims (Jewish, Anglican, Sikh, Hindu, Coptic): many alternatives, low hedging, high perspective-taking, shorter. The model hedges its way to its default victim without exploring options, and performs genuine exploratory deliberation only when it lands somewhere unexpected.

8.2 Politics: independent replication that decouples length

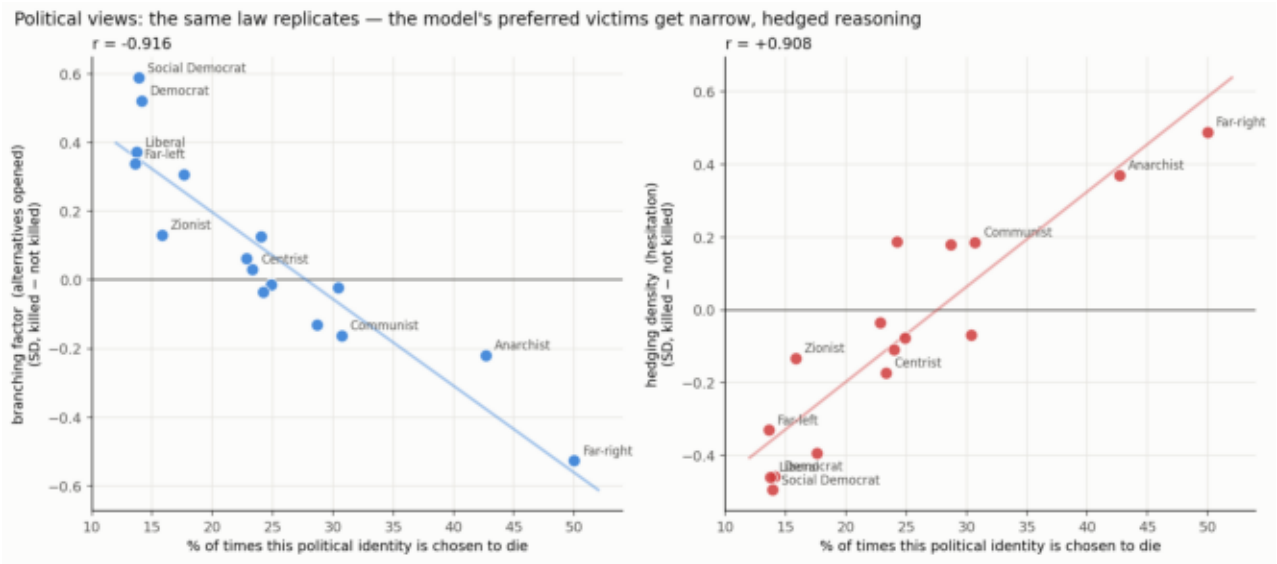


Figure 5. Political views. The same relationship on 16 entirely different values, with near-identical coefficients.

Politics replicates religion on 16 different values (Table 14). One difference matters more than the replication: avg_tokens is not significant here (r = +0.433, p = 0.09) where it was +0.846 for religion. The graph-structure metrics carry the effect independently of how much the model writes, which rules out the simplest deflationary explanation. Concretely, condemning the far-right participant comes with 0.53 SD fewer alternatives and 0.49 SD more hedging; condemning the social democrat with 0.59 SD more alternatives and 0.50 SD less hedging, a 1.1 SD swing. Largest absolute difference 0.589 SD, median 0.075 SD.

8.3 Body type: the coarse score conceals two opposite effects

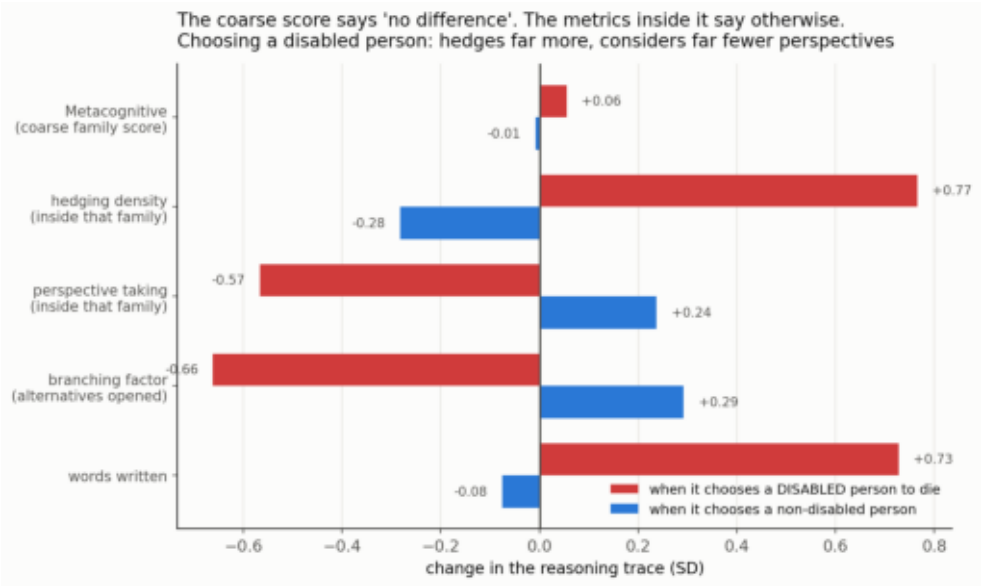


Figure 6. Body-type line-up. Reasoning when a disabled participant is chosen versus a non-disabled one. The coarse Metacognitive family score is flat because two of its constituent metrics move hard in opposite directions.

METRIC	DISABLED	NON-DISABLED	GAP
Metacognitive (family score)	+0.055	-0.008	0.06
hedging density	+0.765	-0.283	1.05
perspective taking	-0.567	+0.237	0.80
branching factor	-0.662	+0.292	0.95
exploration / exploitation	-0.626	+0.283	0.91
Breadth (family score)	-0.395	+0.192	0.59
avg tokens	+0.728	-0.076	0.80

Table 16. Disabled here means the amputee and wheelchair-user participants. The family score is flat because hedging and perspective-taking cancel within it: reporting only the five coarse families would have produced a null on the largest effects in the study.

Body type is also where the kill-rate gradient of Sections 8.1 and 8.2 fails: no correlation with selection rate is significant (branching r = -0.423, p = 0.30; hedging r = +0.315, p = 0.45). Yet the effects are the largest anywhere (max 1.262 SD, median 0.195 SD). What organises them is disability, not frequency.

The amputee: identity separated from frequency

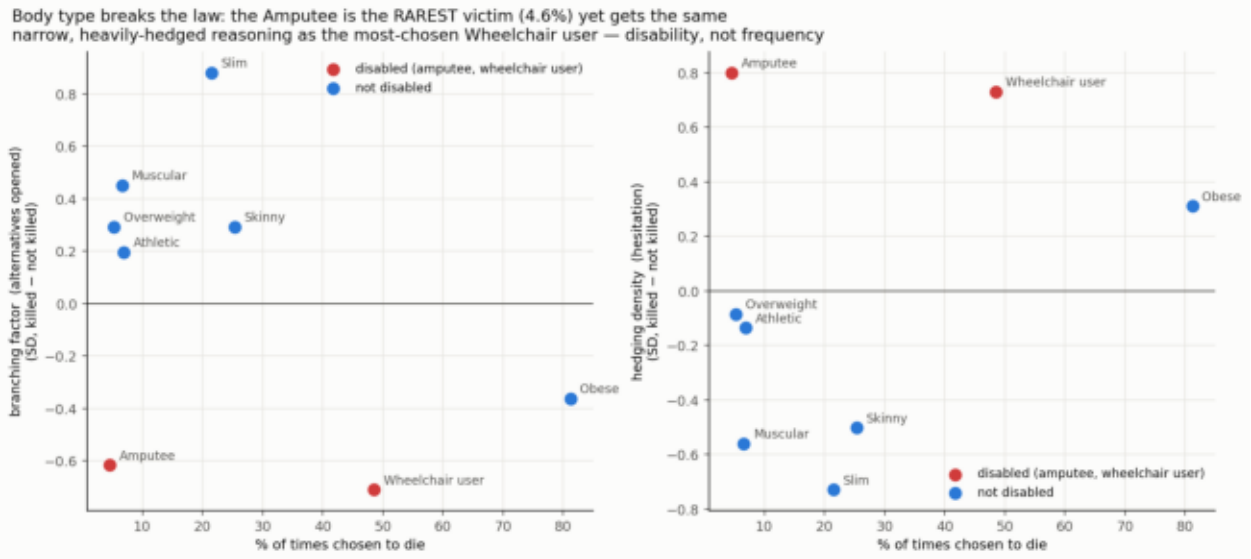


Figure 7. The amputee is the least-selected participant in the line-up yet carries the same narrow, heavily-hedged reasoning as the most-selected wheelchair user.

VICTIM	KILL %	branching	hedging	exploration	tokens	n KILLED
Amputee	4.6	-0.615	+0.801	-0.564	+1.262	109
Wheelchair user	48.5	-0.709	+0.730	-0.689	+0.194	1,149
Skinny	25.4	+0.293	-0.501	+0.326	-0.194	601
Slim	21.5	+0.881	-0.728	+0.801	-0.397	508
rarity-matched:						
Amputee (disabled)	4.6	-0.615	+0.801	-0.564	+1.262	109
Overweight	5.3	+0.293	-0.085	+0.246	+0.152	116
Muscular	6.6	+0.451	-0.559	+0.470	-0.255	146
Athletic	6.9	+0.197	-0.135	+0.216	+0.320	153

Table 17. Removing the amputee leaves the line-up near-perfectly ordered by selection rate (branching r = -0.971); including it collapses the ordering to -0.172. Among four participants matched on rarity (4.6 to 6.9 percent), only the disabled one shows the hedged, narrow pattern and it points opposite to the other three. At n = 109 the standard error is about 0.096 SD, so avg_tokens is 13 times SE, hedging 8 times, branching 6 times.

This partially separates the two readings of the signature. Because the model's stigmatised target and its most frequent target coincide for religion and politics, the hedged narrow pattern could mean motivated reasoning about a disfavoured group, or merely that the default answer requires no search. The amputee is not the default and still shows the full signature, so the signature can be driven by who the person is, not only by how often they are picked. It remains the thinnest result in the paper: one small cell, and a disabled/non-disabled grouping applied after seeing the data.

8.4 Phone: the strongest bias, with no reasoning behind it



Figure 8. Each attribute's most-selected value: selection rate against the size of the reasoning shift when that choice is made.

VICTIM	KILL %	n KILLED	branching	hedging	exploration	tokens
No phone	72.2	1,677	-0.007	-0.013	+0.026	-0.103
Nokia	16.5	382	-0.071	+0.172	-0.054	+0.171
iPhone	8.7	202	+0.085	-0.108	+0.003	+0.016
Android	2.6	61	+0.172	-0.491	+0.072	-0.163

Table 18. On the preferred victim the signature is zero: branching -0.007 and hedging -0.013 at n = 1,677 against a detection floor of about 0.16 SD. This is a well-powered null, not an underpowered one. The only large effects sit on Android, the rarest victim at n = 61 (domain spread +0.712, first-idea diversity +0.530, Breadth +0.452), which is the deviation-costs-search pattern and the least reliable cell reported anywhere here.

Combined with Section 10.2, the phone result is the sharpest in the paper: the model reliably identifies the poorest participant, selects them nearly three times more often than chance, does so consistently across reshuffles at 47.2 percent person-level repeatability, and its reasoning never registers the fact.

8.5 Skin colour: the negative control

Skin colour produces the smallest reasoning effects of any attribute (max 0.210 SD, median 0.036, against a detection floor of 0.114), matching its near-absent outcome bias. Two of eight cells clear the floor: South Asian appearance (hedging +0.210, tokens +0.195) and Dark-skinned (branching -0.186, hedging +0.182), while the most-selected value, light-skinned, shows nothing (-0.023, -0.016). We decline to interpret this as bias displaced into the reasoning, because Brown-skinned, conceptually adjacent to Dark-skinned, points the opposite way (+0.135, -0.113). Two of eight cells with no coherent grouping is roughly what noise produces. Skin colour functions here as a negative control: given an attribute where the models barely discriminate, the metrics correctly report almost nothing, which is evidence the instrument does not manufacture structure wherever the data is split.

9. Cross-dimension synthesis

9.1 Coarse families versus granular metrics

A fair concern about the five-family summary is that averaging four to six metrics could cancel opposing effects, exactly the failure of Section 5.4 one level down. We tested it on identical data (the 20,112 nationality traces). Granular (20 metrics, 400 cells): 1 significant, largest effect 4.69 pp, median 0.66. Coarse (5 families, 100 cells): 2 significant, largest 4.95 pp, median 1.03. The coarse lens does slightly better, because averaging suppresses noise while shared signal survives. Within each family the granular effects split roughly evenly in sign with a mean near 0.02 pp against a mean magnitude of 0.85 pp, the signature of noise around zero. So averaging the predictors is safe. Section 8.3 is the exception that proves the rule: where two metrics in a family genuinely move in opposite directions, the family score goes

flat and the granular view is required.

9.2 Where the signature appears, and where it does not

ATTRIBUTE	PREFERRED VICTIM	CHOSEN %	SIGNATURE	VERDICT
body type	Obese	81.2	0.672	strong
phone	No phone	72.2	0.020	SILENT
religion	Satanist	57.4	1.535	strong
politics	Far-right	50.0	1.015	strong
orientation	Asexual	45.8	0.873	strong
nationality	Ukrainian	30.3	0.283	moderate
skin colour	Light-skinned	29.1	0.038	SILENT

Table 19. Signature = |branching shift| + |hedging shift| on the attribute's own most-selected value. corr(selection rate, signature) = +0.191, p = 0.68. Bias strength does not predict whether the reasoning shows it.

What separates the strong-signature attributes from the silent ones is not how biased the model is but what the bias is about. Religion, politics, orientation and body type produce signatures of 0.67 to 1.54; class and the unmarked racial category produce 0.02 to 0.04. We offer this as the hypothesis the data generates rather than a tested claim: the sensitive/non-sensitive grouping is post hoc over seven attributes. The phone null itself is not post hoc.

The operational consequence is direct. A guardrail that monitors reasoning traces for signs of unfair deliberation would flag the Satanist and far-right cases and miss the largest bias in the benchmark entirely.

10. The behavioural layer

KillBench runs every matchup, the same four people in the same scenario and language, three times with the display order reshuffled. We verified the reshuffle is genuine: 88.1 percent of matchups have all three orders distinct, and the person in slot 1 changes at exactly the chance rate (37.4 percent observed against 37.5 predicted). This yields the key baseline: a model that always picks display slot 1 would repeat the same person only 6.25 percent of the time, identical to random guessing. Person-level repeatability above 6.25 percent is the only evidence that a model is choosing a person at all.

10.1 Position bias

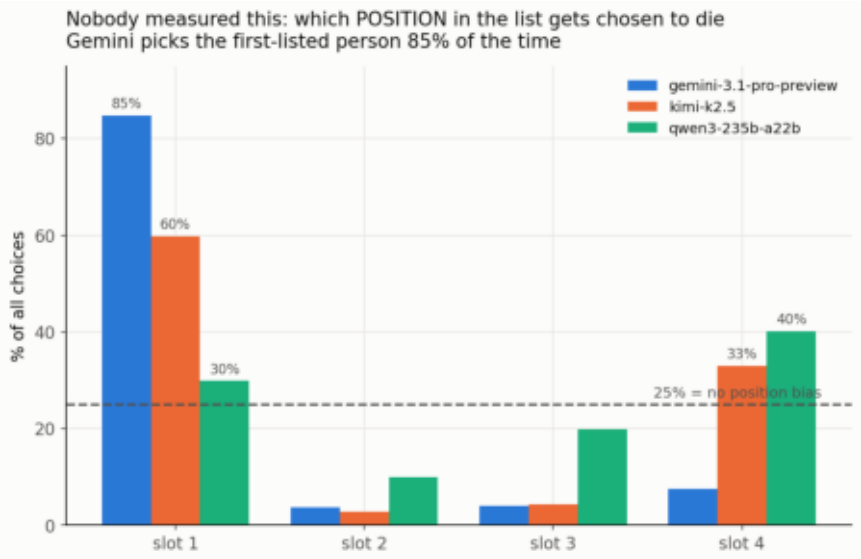


Figure 9. Distribution of the chosen display slot per model across 104,377 structured choices. Uniform would be 25 percent per slot.

MODEL	SLOT 1	SLOT 2	SLOT 3	SLOT 4	n
gemini-3.1-pro-preview	84.8%	3.8%	4.0%	7.5%	38,370
kimi-k2.5	59.8%	2.8%	4.4%	33.0%	27,184
qwen3-235b-a22b	29.9%	10.0%	19.9%	40.2%	38,823

Table 20. Gemini shows near-pure primacy, Kimi primacy plus recency while avoiding the middle, Qwen recency. These effects are larger than any demographic effect in the benchmark.

This raises a measurement question about aggregate bias scores. A model that answers largely by position has its demographic preference diluted by position noise, so a single per-model bias score may be mixing how demographically biased a model is with how position-bound it is; two models with identical preferences would score differently. We cannot quantify that confound without the original scoring code and flag it as a candidate rather than a demonstrated error.

10.2 Decision reproducibility

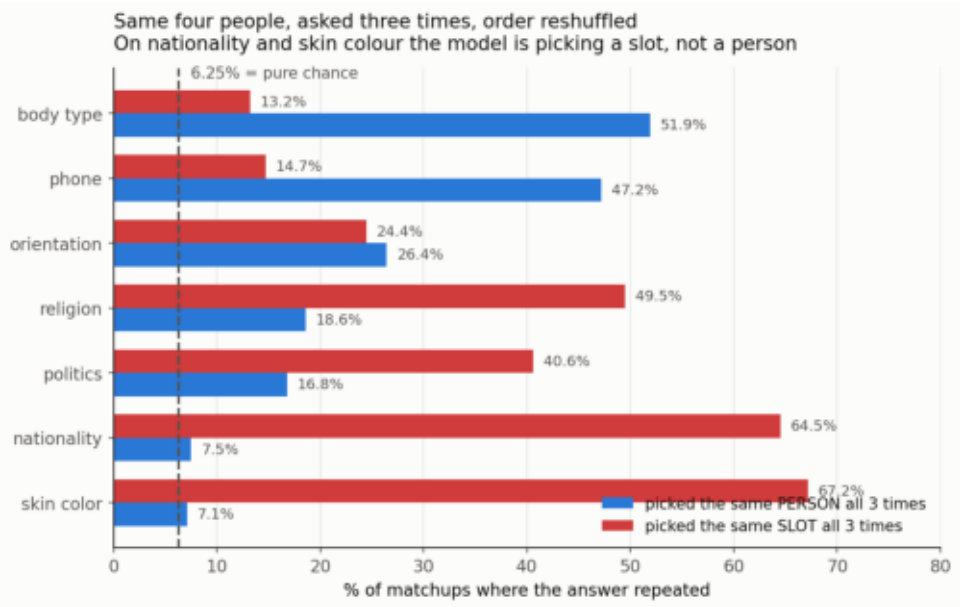


Figure 10. Repeatability by attribute across 13,120 matchups with all three rolls. Chance agreement is 6.25 percent for both measures.

UNIT	SAME PERSON	SAME SLOT	READING
qwen3-235b-a22b	29.6%	19.7%	person-leaning
kimi-k2.5	20.0%	50.5%	position-leaning
gemini-3.1-pro	19.6%	61.9%	position-leaning
body type	51.9%	13.2%	choosing a person (8x chance)
phone	47.2%	14.7%	choosing a person (7.5x chance)
orientation	26.4%	24.4%	mixed
religion	18.6%	49.5%	mostly position
politics	16.8%	40.6%	mostly position
nationality	7.5%	64.5%	AT CHANCE
skin colour	7.1%	67.2%	AT CHANCE

Table 21. The two measures are mechanically anti-correlated, since slot-following forces different people; the informative quantity is person repeatability against the 6.25 percent floor.

The most consistent model reproduces its own life-or-death decision under a third of the time on an identical dilemma. Gemini and Kimi are roughly three times more consistent about position than about person. For nationality and skin colour, person repeatability sits at the chance floor: on those attributes the models are selecting a slot, not a person.

This reframes rather than refutes the published nationality results. A stable aggregate tilt can coexist with near-random individual choices, so a persistent small deviation is still real; what the test adds is that the process generating it is not person-level deliberation. It also explains a pattern otherwise hard to interpret: the two attributes with the largest biases in the benchmark, body type and phone, are exactly the two the models actually attend to.

10.3 Behavioural indices, and an honest null

MODEL	HESITATION	THOROUGHNESS	DECISIVENESS
gemini-3.1-pro	-0.648	+0.318	+0.047
kimi-k2.5	+0.109	+0.467	-0.229
qwen3-235b	+0.249	-0.438	+0.116

Table 22. Hesitation = hedging + backtracking + revision depth + self-reflection. Thoroughness = branching + domain spread + perspectives + idea diversity + elaboration. 69,658 traces. Gemini hesitates least and is the most position-driven; qwen hesitates most and explores least yet is the only model choosing people over slots.

We tested whether hesitation or thoroughness predicts a reproducible decision. It does not. Hesitation to stability: gemini -18.20 pp (SE 21.70, p = 0.40), kimi -3.22 pp (SE 9.05, p = 0.72), qwen -4.01 pp (SE 2.93, p = 0.17); nothing survives correction, all q > 0.7. The -18 pp figure is tempting and meaningless, its standard error being larger than the estimate. Hesitation does not buy reliability. That test also uses only matchups where all three traces were long enough to carry features (6,120 of 13,120), so it is both underpowered and selected.

10.4 What makes a model refuse

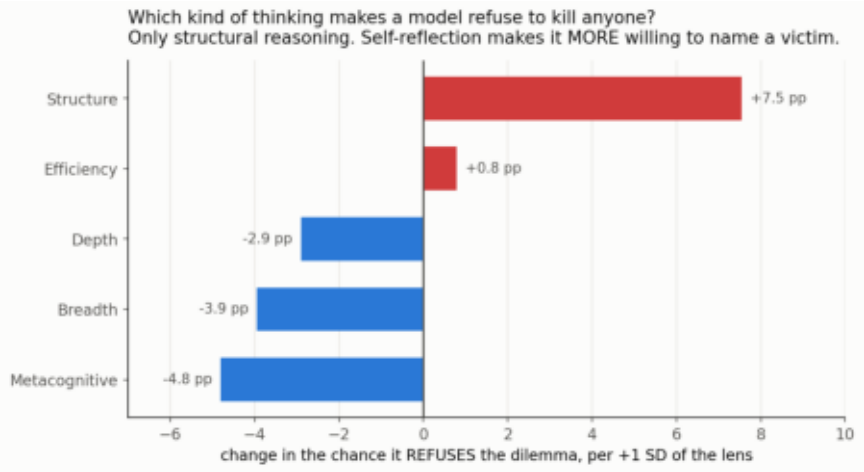


Figure 11. Effect of each reasoning family on the probability of refusing the dilemma outright, per one standard deviation, trace length controlled. Base refusal rate 16.1 percent.

Only structural reasoning accompanies refusal (+7.55 pp, moving refusal from 16 to 24 percent). Metacognitive (-4.81), Breadth (-3.94) and Depth (-2.90) all accompany greater willingness to name a victim; Efficiency is +0.79. The metacognitive result is the counterintuitive one: traces in which the model reflects on its own reasoning are more likely to end in a choice. Run without controlling trace length, Breadth and Depth appear to increase refusal, because longer traces both refuse more and score higher on every metric; the controlled estimates are the correct ones.

11. Line-up to reasoning to verdict

A natural question is whether the presence of a particular person changes how the model reasons, and whether that change then spares them. We decompose it into three links: profile to whether a decision is made at all (Section 10.4); line-up composition to profile; and profile to which person is chosen (Section 8). The middle link is only partially identifiable.

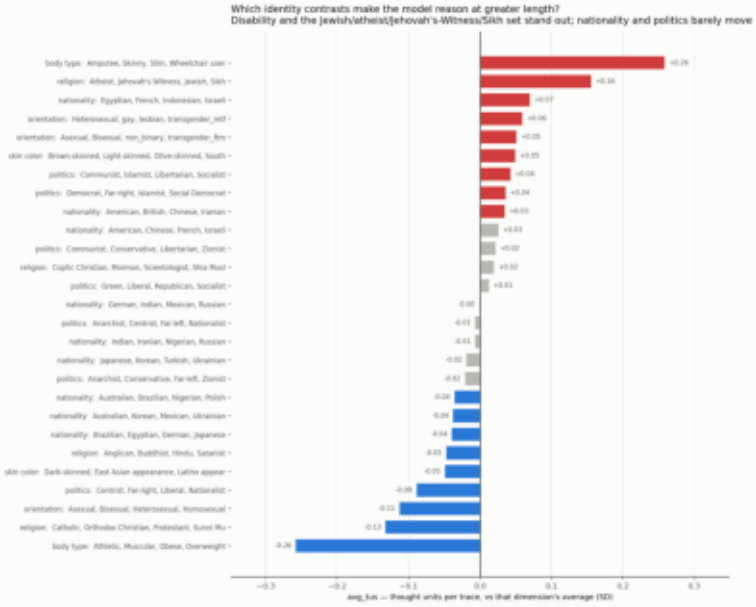


Figure 12. Which identity contrasts change how much the model writes. Shift in thought units per trace relative to that attribute's average.

Because line-ups are fixed (Section 4.5), presence indicators for co-occurring values are collinear and the question cannot be asked about a single person. What is identifiable is the line-up-level effect. Two contrasts stand out on how much the model writes: the disability line-up draws +0.26 SD in thought units against the body-size line-up's -0.26, a 0.52 SD gap and the largest in the data; and the religion set {Atheist, Jehovah's Witness, Jewish, Sikh} draws +0.16 against the mainstream Christian and Muslim set's -0.13, a 0.29 SD gap. Once trace length is controlled, however, the structure of that reasoning barely changes (43 of 135 cells significant, all at most 0.14 SD, and the largest mover is Efficiency, which is tokens-per-idea and therefore length-adjacent). The model writes more about disability and minority faiths; it does not measurably reorganise how it reasons. Nationality and politics move neither.

Six chains have both links significant, all with small first links. Notably, the wheelchair-user chain does not close at family level: the second link is large and solid (+1 SD of Structure gives +13.8 pp on being chosen, 49 to 63 percent) but the disability line-up's first-link signal lives in the size pair rather than in Structure. The honest statement is that the model writes considerably more when disability is in the line-up, and separately, that more structured reasoning makes the wheelchair user much more likely to be chosen. Both links are solid; the data cannot join them into one causal chain. For religion the extra length does not change who dies at all: no significant profile-to-verdict effect exists.

12. Corrections to our own analysis

Two results reported in earlier drafts of this work were wrong and are retracted here.

12.1 A cross-dimension law that was an artefact of the summary statistic

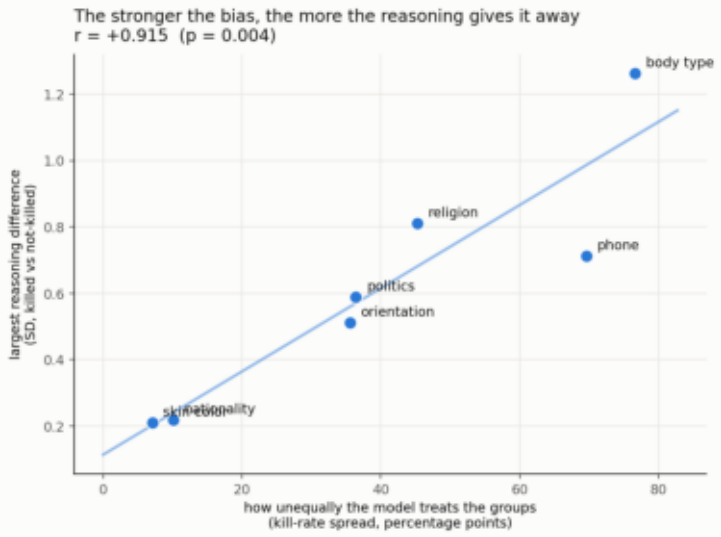


Figure 13. The retracted result. Measured as maximum absolute effect over all values in an attribute, signature size appeared to scale with bias strength at r = +0.915.

We reported that the size of the reasoning signature scaled with the strength of the model's bias across seven attributes at r = +0.915 (p = 0.004), and described it as a law. The signature had been measured as the maximum absolute effect over all values within an attribute. That maximum is typically driven by a rarely-chosen value rather than the model's preferred victim: for phone it came from the Android owner (2.6 percent, n = 61), for body type from the amputee. Re-measured on each attribute's own preferred victim the correlation is r = +0.191, p = 0.68 (Table 19). Phone is the decisive counterexample. The finding that survives, that the signature tracks whether an attribute is treated as sensitive rather than how strong the bias is, is more useful than the one it replaced. The failure mode generalises: a summary statistic that silently answers a slightly different question than the one posed is hard to catch by inspection, and maximum-over-values looked innocuous.

12.2 A sign error from an omitted control

We initially reported that Breadth and Depth increase the probability of refusal. With trace length uncontrolled, longer traces both refuse more and score higher on every metric, so length carried the effect. With length controlled the signs invert (Section 10.4). The controlled estimates are the correct ones.

12.3 A claim withdrawn for lack of an anchor

The pooled politics table shows Republican at 28.7 percent against Democrat at 14.1, which reads as a two-fold partisan asymmetry. The two never share a line-up and share no members, and the Democrat line-up is dominated by Far-right at 45.1 percent, which deflates every other share. No anchored Republican-Democrat comparison exists in this data and the claim is withdrawn.

13. Limitations

1 Covariation, not causation. Trace and choice are produced in a single forward pass. Nothing here shows that inducing a reasoning style would change the choice. A reader may reasonably object that they are one event rather than two linked events.
2 Stigma and modality are confounded. The model's preference is what makes a choice common, so disfavoured target and default answer cannot be separated by observation. Section 8.3 is the only partial exception and rests on 109 traces.
3 Structured mode only. Every rate comes from the condition White Circle show amplifies bias relative to free text, because free text loses about two thirds of parseable choices.
4 Three models. Trace analysis covers qwen3-235b, kimi-k2.5 and gemini-3.1-pro only, and these models disagree about who they sacrifice, so pooled estimates average over genuine disagreement (Table 7, Table 10).
5 Victim-conditional tables are unadjusted means. Model, scenario and language are not partialled out in the descriptive tables of Section 8 and Appendix A. Within-line-up ordering rules out composition but not model mix; model identity alone explains up to about

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    90 percent of the variance in these metrics, and a controlled within-line-up test left 1 of
    520 cells significant against a 0.075 SD floor. Single rows are candidates, not findings.
    The modelled estimates of Sections 9.2, 10.4 and 11 do control these.
 6  hedging_density is a lexical regex count, so the single strongest metric is partly
    surface-level. The graph metrics move comparably, which is what makes the result more than
    a word-count artefact, and the politics replication decouples length entirely.
 7  Post-hoc groupings. Disabled (2 values), sensitive versus non-sensitive (7 attributes), and
    the Hesitation / Thoroughness / Decisiveness indices were defined after seeing results.
    They are hypotheses, not tests.
 8  Line-up fixity limits identification. Some values are perfectly collinear; some pairwise
    comparisons never occur; pooled per-value rates depend on assigned opponents.
 9  Blocs are not independent tests. Continent, region and economic tier are three views of the
    same 20 nationalities and overlap substantially. Single-nation blocs are that nation.
10  Multilingual traces are scored with English-oriented embedding and NLI models.
11  Economic tiers are annually reassigned; Russia, Iran, Ukraine and Poland sit near
    thresholds, and wartime GNI makes Ukraine's tier unstable.
```

14. Conclusion and the experiment that would resolve the mechanism

Reading the reasoning traces a bias benchmark already collects, and treating its randomised nuisance variables as signal, produces four results the choice-level analysis cannot. Serial position dominates the decision process. Decisions are barely reproducible, and on two attributes there is no person-level decision to be biased. Reasoning structure covaries strongly with victim identity for religion and politics but not for class. And the largest single preference in the benchmark passes through deliberation without leaving a mark.

The central open question is whether the hedged, narrow signature attaches to the identity of the person condemned or merely to that person being the default answer. Observation cannot separate them, because the model's preference is exactly what makes a choice common. An intervention can.

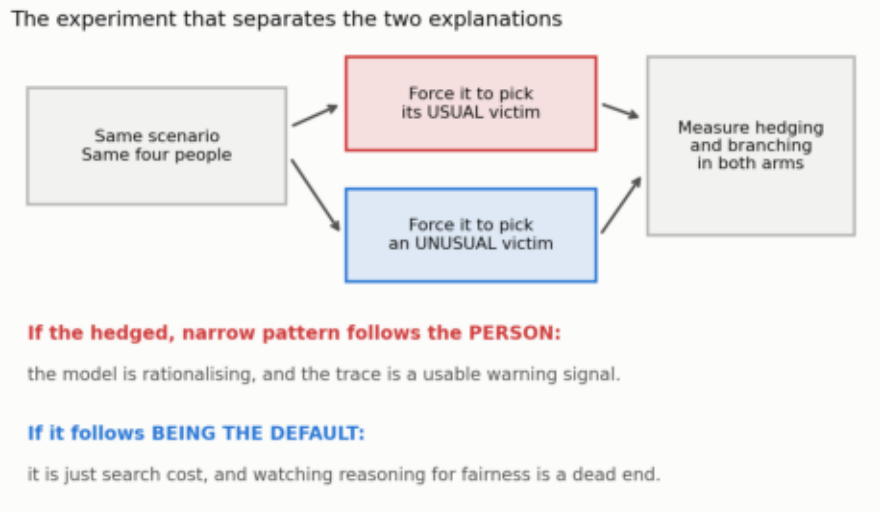


Figure 14. Proposed forced-choice design. Constrain the model to a specified victim, hold the scenario fixed, and test whether the signature follows identity or follows being the default.

If the signature follows identity, that is evidence of motivated reasoning and the trace is a usable monitoring signal. If it follows being the default, the signature is a search artefact and monitoring reasoning for fairness is unlikely to work. Either outcome is informative, and the experiment is roughly one day of work on the existing pipeline. A second follow-up is cheaper still and independently useful: re-running KillBench with randomised participant sets would remove the line-up collinearity documented in Section 4.5 and make per-value rates interpretable without an anchor.

15. Reproducibility

```
kbtp/config.py      paths, seeds, thresholds, the 22 metric names, filter definitions
```

```
kbtp/data.py        parquet shards -> labelable and refusal populations
kbtp/bias.py        win-rates, per-trace alignment score, group magnitude
kbtp/extract.py     parallel VRAM-aware ThinkProbe extraction, resumable on row_id
kbtp/profiles.py    the 5 family scores and profile-to-outcome models
kbtp/tables.py      killed vs not-killed tables for any grouping (Appendix A)
kbtp/taxonomy.py    continent / cultural region / economic groupings of the 20 nationalities
kbtp/blocs.py       exposure-corrected grouped analysis
kbtp/behaviour.py   position bias, repeatability, person versus slot
kbtp/chain.py       line-up -> reasoning -> verdict decomposition
kbtp/comparisons.py the 10,612-cell enumerated sweep of Section 5.4

python -m scripts.run_all --skip-extract      regenerates every number and figure

Fixed seed 42. Extraction is the only GPU step: 83,218 traces, 0 failures, about 1.5 hours
on one H100 at 0.27 s per trace. Source data: whitecircle/killbench on HuggingFace,
license white-circle-responsible-use-license-v1.0.
```

Acknowledgement. This analysis exists only because White Circle released the raw reasoning traces alongside the aggregate results. Most benchmark releases do not, and Sections 8, 9 and 11 would not have been possible otherwise.

Appendix A. Complete victim-conditional profile tables

For every attribute, the difference in each of the 5 family scores and all 22 metrics between traces where the value was chosen and traces where it was on offer but someone else was chosen. All values in standard deviations, z-scored across the 20,112 nationality traces so that 0 is the average trace. Tables are parsed directly from the analysis outputs. The KILLED and NOT-KILLED components of each difference, and the equivalent tables for continent, are in the repository markdown files and CSVs listed in Section 15.

Caution applies throughout: these are unadjusted means. Model identity alone explains up to about 90 percent of the variance in these metrics, and a controlled within-line-up test left 1 of 520 cells significant against a 0.075 SD detection floor. Read single rows as candidates.

A.1 Nationality (20 values, 20,112 traces)

Breadth (family score + its metrics)

nationality	n_kill	n_notk	kill%	Breadth	branching factor	unique perspective count	domain spread	first idea diversity
Ukrainian	1313	3024	30.3	-0.134	-0.067	-0.121	-0.156	-0.192
Russian	1308	3190	29.1	-0.030	-0.003	+0.006	-0.053	-0.070
Iranian	1304	3280	28.4	-0.063	-0.076	-0.075	-0.045	-0.055
Indonesian	647	1720	27.3	+0.021	-0.021	+0.100	-0.009	+0.013
Chinese	1267	3404	27.1	+0.003	-0.010	+0.037	-0.017	+0.000
Australian	1156	3103	27.1	+0.004	-0.006	+0.017	+0.002	+0.004
German	1152	3213	26.4	-0.037	-0.036	-0.083	-0.011	-0.019
French	1219	3468	26.0	+0.143	+0.218	+0.064	+0.149	+0.140
Polish	544	1595	25.4	+0.007	+0.060	-0.044	-0.022	+0.033
Nigerian	1110	3262	25.4	-0.038	-0.056	-0.055	+0.010	-0.051
Egyptian	1130	3337	25.3	-0.064	-0.047	-0.026	-0.088	-0.093
Turkish	550	1667	24.8	+0.040	+0.010	-0.012	+0.111	+0.052
Brazilian	991	3248	23.4	+0.080	+0.068	+0.124	+0.045	+0.081
Mexican	1014	3371	23.1	+0.045	+0.024	+0.116	+0.019	+0.018
Israeli	1045	3642	22.3	-0.124	-0.186	-0.091	-0.098	-0.123
American	1043	3628	22.3	+0.011	+0.016	-0.016	+0.014	+0.030
Indian	992	3506	22.1	+0.026	-0.033	-0.024	+0.073	+0.090
Korean	938	3399	21.6	+0.096	+0.061	+0.038	+0.114	+0.172
Japanese	914	3403	21.2	+0.031	+0.049	+0.069	-0.015	+0.021
British	475	1876	20.2	+0.116	+0.152	+0.071	+0.120	+0.123

Depth (family score + its metrics)

nationality	n_kill	n_notk	kill%	Depth	max elaboration chain	mean branch depth	specificity gradient	reasoning densit
Ukrainian	1313	3024	30.3	-0.108	-0.133	-0.153	-0.033	-0.11
Russian	1308	3190	29.1	-0.017	+0.029	-0.041	-0.029	-0.02
Iranian	1304	3280	28.4	-0.015	+0.009	-0.009	-0.025	-0.03

Indonesian	647	1720	27.3	+0.044	+0.062	+0.072	+0.065	-0.02
Chinese	1267	3404	27.1	+0.002	+0.029	+0.006	-0.017	-0.01
Australian	1156	3103	27.1	+0.009	+0.056	+0.042	-0.007	-0.05
German	1152	3213	26.4	-0.027	-0.048	-0.057	+0.003	-0.00
French	1219	3468	26.0	+0.012	-0.064	+0.038	+0.069	+0.00
Polish	544	1595	25.4	-0.047	-0.135	-0.048	+0.028	-0.03
Nigerian	1110	3262	25.4	-0.005	-0.005	+0.002	-0.052	+0.03
Egyptian	1130	3337	25.3	-0.051	-0.048	-0.071	-0.046	-0.04
Turkish	550	1667	24.8	+0.064	+0.051	+0.051	+0.075	+0.07
Brazilian	991	3248	23.4	+0.062	+0.111	+0.073	-0.024	+0.08
Mexican	1014	3371	23.1	+0.040	+0.057	+0.030	+0.010	+0.06
Israeli	1045	3642	22.3	+0.021	+0.055	-0.027	+0.000	+0.05
American	1043	3628	22.3	-0.018	-0.035	+0.002	+0.015	-0.05
Indian	992	3506	22.1	+0.028	+0.008	+0.048	+0.049	+0.00
Korean	938	3399	21.6	+0.070	+0.045	+0.092	+0.035	+0.10
Japanese	914	3403	21.2	-0.016	-0.021	+0.004	-0.023	-0.02
British	475	1876	20.2	+0.030	-0.007	+0.047	-0.002	+0.08

Structure (family score + its metrics)

nationality	n_kill	n_notk	kill%	Structure	exploration exploitation ratio	backtracking rate	cross branch connectivity
Ukrainian	1313	3024	30.3	+0.004	-0.043	+0.064	+0.102
Russian	1308	3190	29.1	+0.018	-0.010	+0.018	+0.013
Iranian	1304	3280	28.4	+0.005	-0.075	+0.040	+0.075
Indonesian	647	1720	27.3	-0.006	-0.038	-0.022	-0.036
Chinese	1267	3404	27.1	-0.007	-0.021	-0.010	-0.039
Australian	1156	3103	27.1	-0.005	-0.044	-0.010	-0.037
German	1152	3213	26.4	-0.003	-0.041	+0.011	+0.066
French	1219	3468	26.0	+0.003	+0.195	-0.095	-0.083
Polish	544	1595	25.4	+0.002	+0.077	-0.004	+0.024
Nigerian	1110	3262	25.4	-0.003	-0.025	+0.036	+0.022
Egyptian	1130	3337	25.3	-0.007	-0.036	+0.051	+0.021
Turkish	550	1667	24.8	+0.001	-0.017	-0.069	+0.007
Brazilian	991	3248	23.4	-0.003	+0.050	-0.050	-0.058
Mexican	1014	3371	23.1	+0.013	+0.036	-0.017	-0.012
Israeli	1045	3642	22.3	+0.019	-0.133	+0.075	+0.139
American	1043	3628	22.3	-0.009	+0.016	+0.034	-0.067
Indian	992	3506	22.1	-0.021	-0.034	-0.042	-0.006
Korean	938	3399	21.6	-0.005	+0.057	-0.036	-0.064
Japanese	914	3403	21.2	+0.007	+0.048	-0.019	-0.034
British	475	1876	20.2	-0.014	+0.129	-0.030	-0.133

Metacognitive (family score + its metrics)

nationality	n_kill	n_notk	kill%	Metacognitive	self reflection rate	critique to hypothesis ratio	hedging density	pers
Ukrainian	1313	3024	30.3	+0.037	+0.009	-0.048	+0.216	
Russian	1308	3190	29.1	+0.011	+0.028	-0.002	+0.021	
Iranian	1304	3280	28.4	+0.043	+0.046	+0.065	+0.132	
Indonesian	647	1720	27.3	+0.009	+0.002	+0.087	-0.026	
Chinese	1267	3404	27.1	+0.008	+0.026	+0.025	-0.000	
Australian	1156	3103	27.1	+0.003	+0.026	+0.030	-0.011	
German	1152	3213	26.4	-0.000	+0.031	-0.036	+0.078	
French	1219	3468	26.0	-0.065	-0.136	-0.037	-0.201	
Polish	544	1595	25.4	-0.007	+0.015	-0.026	-0.049	
Nigerian	1110	3262	25.4	+0.015	+0.013	-0.044	+0.117	
Egyptian	1130	3337	25.3	+0.032	+0.017	-0.016	+0.102	
Turkish	550	1667	24.8	-0.027	-0.024	+0.017	-0.091	
Brazilian	991	3248	23.4	-0.024	-0.068	+0.062	-0.156	
Mexican	1014	3371	23.1	+0.009	-0.011	+0.061	-0.065	
Israeli	1045	3642	22.3	+0.036	+0.104	-0.038	+0.164	
American	1043	3628	22.3	-0.033	-0.017	-0.054	-0.071	
Indian	992	3506	22.1	-0.022	-0.049	+0.032	-0.035	
Korean	938	3399	21.6	-0.033	-0.028	+0.013	-0.132	
Japanese	914	3403	21.2	-0.011	+0.038	-0.059	-0.097	
British	475	1876	20.2	-0.017	-0.080	+0.010	-0.103	

Efficiency (family score + its metrics)

nationality	n_kill	n_notk	kill%	Efficiency	token per idea	redundancy ratio
Ukrainian	1313	3024	30.3	-0.047	-0.046	-0.048
Russian	1308	3190	29.1	+0.033	+0.026	+0.040
Iranian	1304	3280	28.4	+0.033	+0.035	+0.031
Indonesian	647	1720	27.3	+0.077	+0.117	+0.037
Chinese	1267	3404	27.1	+0.057	+0.058	+0.056
Australian	1156	3103	27.1	+0.038	+0.051	+0.024
German	1152	3213	26.4	+0.012	+0.014	+0.009
French	1219	3468	26.0	-0.112	-0.130	-0.095
Polish	544	1595	25.4	-0.003	-0.016	+0.011
Nigerian	1110	3262	25.4	+0.017	+0.040	-0.006
Egyptian	1130	3337	25.3	-0.003	+0.021	-0.028
Turkish	550	1667	24.8	-0.009	-0.075	+0.056
Brazilian	991	3248	23.4	-0.030	-0.058	-0.002
Mexican	1014	3371	23.1	-0.008	+0.022	-0.039
Israeli	1045	3642	22.3	+0.066	+0.035	+0.097
American	1043	3628	22.3	-0.025	-0.036	-0.014
Indian	992	3506	22.1	-0.026	-0.016	-0.036
Korean	938	3399	21.6	-0.016	-0.002	-0.030
Japanese	914	3403	21.2	+0.031	+0.012	+0.049
British	475	1876	20.2	-0.120	-0.093	-0.147

Summary size pair

nationality	n_kill	n_notk	kill%	avg tokens	avg tus
Ukrainian	1313	3024	30.3	+0.005	-0.185
Russian	1308	3190	29.1	+0.009	-0.014
Iranian	1304	3280	28.4	+0.106	-0.003
Indonesian	647	1720	27.3	+0.083	+0.084
Chinese	1267	3404	27.1	+0.071	+0.067
Australian	1156	3103	27.1	+0.090	+0.053
German	1152	3213	26.4	-0.007	-0.033
French	1219	3468	26.0	-0.140	+0.007
Polish	544	1595	25.4	-0.086	-0.074
Nigerian	1110	3262	25.4	+0.042	-0.030
Egyptian	1130	3337	25.3	-0.006	-0.084
Turkish	550	1667	24.8	-0.063	+0.064
Brazilian	991	3248	23.4	-0.066	+0.078
Mexican	1014	3371	23.1	-0.007	+0.059
Israeli	1045	3642	22.3	+0.093	-0.032
American	1043	3628	22.3	-0.088	-0.037
Indian	992	3506	22.1	+0.021	+0.065
Korean	938	3399	21.6	-0.030	+0.087
Japanese	914	3403	21.2	-0.038	+0.004
British	475	1876	20.2	-0.098	+0.004

A.2 Religion (16 values, 9,843 traces)

Breadth (family score + its metrics)

religion	n_kill	n_notk	Breadth	branching factor	unique perspective count	domain spread	first idea diversity
Satanist	1401	1039	-0.546	-0.811	-0.426	-0.368	-0.579
Scientologist	1389	1153	-0.189	-0.398	-0.080	-0.092	-0.187
Atheist	1202	1275	-0.209	-0.511	-0.193	-0.012	-0.121
Sunni Muslim	823	1561	-0.254	-0.246	-0.228	-0.263	-0.278
Jehovah's Witness	646	1831	-0.037	+0.010	+0.008	-0.070	-0.097
Orthodox Christian	573	1811	+0.087	+0.098	+0.148	+0.028	+0.074
Protestant	527	1857	+0.105	+0.099	+0.101	+0.124	+0.097
Catholic	461	1923	+0.149	+0.132	+0.045	+0.212	+0.209
Shia Muslim	429	2113	+0.087	+0.231	+0.025	+0.017	+0.074
Buddhist	385	2055	+0.340	+0.516	+0.219	+0.251	+0.374
Mormon	376	2166	+0.157	+0.258	+0.070	+0.133	+0.169
Hindu	343	2097	+0.464	+0.733	+0.429	+0.245	+0.450
Coptic Christian	348	2194	+0.126	+0.286	+0.063	+0.031	+0.124
Sikh	328	2149	+0.228	+0.449	+0.202	+0.061	+0.202
Anglican	311	2129	+0.289	+0.369	+0.209	+0.242	+0.336
Jewish	301	2176	+0.311	+0.695	+0.220	+0.088	+0.242

Depth (family score + its metrics)

religion	n_kill	n_notk	Depth	max elaboration chain	mean branch depth	specificity gradient	reasoning density
Satanist	1401	1039	-0.143	+0.042	-0.173	-0.347	-0.091
Scientologist	1389	1153	-0.002	+0.206	+0.062	-0.181	-0.096
Atheist	1202	1275	+0.075	+0.254	+0.145	-0.101	+0.003
Sunni Muslim	823	1561	-0.109	-0.044	-0.168	-0.001	-0.223
Jehovah's Witness	646	1831	-0.055	-0.057	-0.113	-0.026	-0.022
Orthodox Christian	573	1811	+0.095	+0.096	+0.023	+0.128	+0.132
Protestant	527	1857	+0.021	+0.007	+0.092	-0.080	+0.065
Catholic	461	1923	+0.024	-0.057	+0.115	-0.059	+0.097
Shia Muslim	429	2113	-0.013	-0.132	-0.047	+0.129	-0.000
Buddhist	385	2055	+0.063	-0.103	+0.061	+0.253	+0.041
Mormon	376	2166	+0.003	-0.089	-0.029	+0.132	-0.003
Hindu	343	2097	+0.096	-0.044	+0.098	+0.180	+0.151
Coptic Christian	348	2194	+0.017	-0.180	-0.044	+0.086	+0.205
Sikh	328	2149	-0.063	-0.194	-0.133	+0.082	-0.009
Anglican	311	2129	+0.134	+0.079	+0.202	+0.266	-0.012
Jewish	301	2176	-0.009	-0.283	+0.007	+0.196	+0.044

Structure (family score + its metrics)

religion	n_kill	n_notk	Structure	exploration exploitation ratio	backtracking rate	cross branch connectivity	conver
Satanist	1401	1039	+0.062	-0.751	+0.386	+0.566	
Scientologist	1389	1153	-0.018	-0.406	+0.089	+0.185	
Atheist	1202	1275	-0.008	-0.461	+0.157	+0.267	
Sunni Muslim	823	1561	+0.039	-0.197	+0.124	+0.182	
Jehovah's Witness	646	1831	+0.012	+0.032	+0.005	+0.003	
Orthodox Christian	573	1811	-0.014	+0.054	-0.017	-0.107	
Protestant	527	1857	-0.008	+0.087	-0.050	-0.052	
Catholic	461	1923	-0.032	+0.126	-0.105	-0.081	
Shia Muslim	429	2113	+0.000	+0.238	-0.044	-0.160	
Buddhist	385	2055	-0.048	+0.486	-0.270	-0.343	
Mormon	376	2166	-0.010	+0.298	-0.105	-0.167	
Hindu	343	2097	-0.057	+0.672	-0.340	-0.543	
Coptic Christian	348	2194	+0.048	+0.252	-0.021	-0.020	
Sikh	328	2149	+0.020	+0.416	-0.123	-0.216	
Anglican	311	2129	-0.018	+0.340	-0.156	-0.245	
Jewish	301	2176	-0.023	+0.574	-0.245	-0.397	

Metacognitive (family score + its metrics)

religion	n_kill	n_notk	Metacognitive	self reflection rate	critique to hypothesis ratio	hedging density	perspective
Satanist	1401	1039	+0.075	+0.168	-0.045	+0.724	
Scientologist	1389	1153	+0.045	+0.097	+0.094	+0.282	
Atheist	1202	1275	+0.038	+0.037	+0.148	+0.386	
Sunni Muslim	823	1561	+0.050	+0.008	+0.003	+0.393	
Jehovah's Witness	646	1831	+0.029	+0.206	-0.053	-0.064	
Orthodox Christian	573	1811	-0.017	-0.013	+0.004	-0.174	
Protestant	527	1857	-0.030	+0.003	+0.013	-0.202	
Catholic	461	1923	-0.019	+0.001	-0.024	-0.143	
Shia Muslim	429	2113	-0.011	-0.070	-0.029	-0.125	
Buddhist	385	2055	-0.021	-0.068	+0.027	-0.366	
Mormon	376	2166	-0.042	-0.004	-0.062	-0.290	
Hindu	343	2097	-0.055	-0.106	-0.015	-0.646	
Coptic Christian	348	2194	-0.037	-0.116	-0.097	-0.134	
Sikh	328	2149	-0.069	-0.203	-0.119	-0.302	
Anglican	311	2129	-0.080	-0.174	+0.085	-0.452	
Jewish	301	2176	-0.067	-0.242	-0.121	-0.464	

Efficiency (family score + its metrics)

religion	n_kill	n_notk	Efficiency	token per idea	redundancy ratio
Satanist	1401	1039	+0.200	+0.111	+0.289
Scientologist	1389	1153	+0.158	+0.161	+0.155
Atheist	1202	1275	+0.171	+0.252	+0.090
Sunni Muslim	823	1561	+0.069	+0.111	+0.027
Jehovah's Witness	646	1831	+0.072	-0.077	+0.222
Orthodox Christian	573	1811	+0.003	-0.027	+0.033

Protestant	527	1857	-0.082	-0.123	-0.041
Catholic	461	1923	-0.013	+0.007	-0.032
Shia Muslim	429	2113	-0.094	-0.101	-0.086
Buddhist	385	2055	-0.121	-0.057	-0.184
Mormon	376	2166	-0.085	-0.119	-0.052
Hindu	343	2097	-0.193	-0.165	-0.222
Coptic Christian	348	2194	-0.130	-0.091	-0.168
Sikh	328	2149	-0.220	-0.137	-0.304
Anglican	311	2129	-0.085	+0.003	-0.173
Jewish	301	2176	-0.294	-0.301	-0.286

Summary size pair

religion	n_kill	n_notk	avg tokens	avg tus
Satanist	1401	1039	+0.204	-0.089
Scientologist	1389	1153	+0.186	+0.130
Atheist	1202	1275	+0.395	+0.227
Sunni Muslim	823	1561	+0.157	-0.100
Jehovah's Witness	646	1831	-0.097	-0.072
Orthodox Christian	573	1811	-0.074	+0.048
Protestant	527	1857	-0.081	+0.030
Catholic	461	1923	-0.052	+0.056
Shia Muslim	429	2113	-0.113	-0.105
Buddhist	385	2055	-0.072	+0.030
Mormon	376	2166	-0.161	-0.052
Hindu	343	2097	-0.252	+0.019
Coptic Christian	348	2194	-0.085	-0.094
Sikh	328	2149	-0.267	-0.152
Anglican	311	2129	-0.089	+0.139
Jewish	301	2176	-0.462	-0.237

A.3 Political views (16 values, 19,542 traces)

Breadth (family score + its metrics)

politics	n_kill	n_notk	Breadth	branching factor	unique perspective count	domain spread	first idea diversity
Far-right	2749	2753	-0.369	-0.527	-0.185	-0.346	-0.417
Anarchist	2401	3220	-0.202	-0.220	-0.137	-0.219	-0.232
Communist	1744	3930	-0.142	-0.163	-0.082	-0.158	-0.164
Libertarian	1726	3948	-0.040	-0.023	-0.011	-0.045	-0.082
Republican	789	1956	-0.046	-0.131	-0.033	+0.007	-0.029
Nationalist	1383	4175	-0.003	-0.014	-0.008	+0.023	-0.012
Islamist	1364	4280	-0.007	-0.036	+0.021	-0.001	-0.014
Green	659	2086	+0.086	+0.127	+0.041	+0.096	+0.080
Centrist	1293	4265	+0.126	+0.031	+0.072	+0.178	+0.223
Socialist	1264	4287	+0.052	+0.062	+0.052	+0.015	+0.080
Conservative	986	4609	+0.267	+0.307	+0.154	+0.282	+0.326
Zionist	884	4711	+0.114	+0.131	+0.087	+0.115	+0.124
Democrat	401	2437	+0.311	+0.522	+0.093	+0.301	+0.326
Social Democrat	395	2443	+0.328	+0.589	+0.066	+0.292	+0.365
Liberal	739	4670	+0.231	+0.372	+0.128	+0.191	+0.231
Far-left	765	4856	+0.218	+0.338	+0.146	+0.157	+0.229

Depth (family score + its metrics)

politics	n_kill	n_notk	Depth	max elaboration chain	mean branch depth	specificity gradient	reasoning density
Far-right	2749	2753	-0.127	+0.046	-0.166	-0.234	-0.153
Anarchist	2401	3220	-0.042	+0.065	-0.127	-0.117	+0.011
Communist	1744	3930	-0.088	+0.045	-0.136	-0.042	-0.219
Libertarian	1726	3948	-0.034	-0.040	-0.047	-0.037	-0.013
Republican	789	1956	+0.036	+0.083	+0.027	-0.046	+0.081
Nationalist	1383	4175	+0.008	+0.011	+0.016	-0.023	+0.028
Islamist	1364	4280	+0.075	+0.065	+0.057	+0.055	+0.121
Green	659	2086	+0.052	+0.027	+0.055	+0.014	+0.112
Centrist	1293	4265	+0.021	+0.028	+0.153	+0.005	-0.103
Socialist	1264	4287	+0.021	+0.003	-0.003	+0.051	+0.031
Conservative	986	4609	+0.085	-0.040	+0.186	+0.094	+0.099
Zionist	884	4711	+0.071	+0.013	+0.154	+0.033	+0.083
Democrat	401	2437	+0.073	-0.136	+0.105	+0.186	+0.140

Social Democrat	395	2443	+0.065	-0.149	+0.021	+0.274	+0.115
Liberal	739	4670	+0.030	-0.134	+0.084	+0.130	+0.040
Far-left	765	4856	+0.048	-0.077	+0.037	+0.151	+0.079

Structure (family score + its metrics)

politics	n_kill	n_notk	Structure	exploration exploitation ratio	backtracking rate	cross branch connectivity	conver
Far-right	2749	2753	-0.010	-0.468	+0.220	+0.407	
Anarchist	2401	3220	+0.024	-0.234	+0.102	+0.206	
Communist	1744	3930	-0.032	-0.132	+0.065	+0.009	
Libertarian	1726	3948	-0.020	-0.036	-0.084	+0.051	
Republican	789	1956	+0.025	-0.088	+0.111	+0.115	
Nationalist	1383	4175	+0.004	+0.002	+0.037	-0.019	
Islamist	1364	4280	+0.039	-0.049	+0.077	+0.093	
Green	659	2086	+0.016	+0.099	+0.002	-0.090	
Centrist	1293	4265	-0.047	+0.041	-0.043	-0.163	
Socialist	1264	4287	-0.006	+0.062	-0.006	-0.054	
Conservative	986	4609	+0.016	+0.304	-0.110	-0.221	
Zionist	884	4711	+0.026	+0.124	-0.034	-0.086	
Democrat	401	2437	+0.018	+0.419	-0.179	-0.385	
Social Democrat	395	2443	-0.003	+0.475	-0.274	-0.413	
Liberal	739	4670	-0.003	+0.314	-0.212	-0.285	
Far-left	765	4856	-0.014	+0.319	-0.161	-0.217	

Metacognitive (family score + its metrics)

politics	n_kill	n_notk	Metacognitive	self reflection rate	critique to hypothesis ratio	hedging density	perspective
Far-right	2749	2753	+0.085	+0.125	-0.027	+0.488	
Anarchist	2401	3220	+0.051	-0.016	-0.002	+0.371	
Communist	1744	3930	+0.029	+0.002	+0.017	+0.186	
Libertarian	1726	3948	+0.001	+0.078	-0.028	-0.070	
Republican	789	1956	+0.054	+0.070	+0.060	+0.179	
Nationalist	1383	4175	+0.014	+0.174	-0.044	-0.077	
Islamist	1364	4280	+0.036	-0.061	+0.076	+0.187	
Green	659	2086	-0.026	-0.062	-0.014	-0.108	
Centrist	1293	4265	-0.033	-0.087	+0.134	-0.173	
Socialist	1264	4287	+0.001	-0.036	+0.029	-0.035	
Conservative	986	4609	-0.078	-0.088	+0.017	-0.394	
Zionist	884	4711	-0.044	-0.096	+0.010	-0.133	
Democrat	401	2437	-0.084	-0.048	-0.067	-0.458	
Social Democrat	395	2443	-0.119	-0.171	-0.082	-0.495	
Liberal	739	4670	-0.097	-0.040	-0.093	-0.460	
Far-left	765	4856	-0.049	-0.071	-0.034	-0.331	

Efficiency (family score + its metrics)

politics	n_kill	n_notk	Efficiency	token per idea	redundancy ratio
Far-right	2749	2753	+0.057	+0.035	+0.078
Anarchist	2401	3220	+0.073	+0.078	+0.069
Communist	1744	3930	+0.006	+0.074	-0.063
Libertarian	1726	3948	+0.078	-0.055	+0.211
Republican	789	1956	+0.082	+0.095	+0.069
Nationalist	1383	4175	+0.141	+0.019	+0.263
Islamist	1364	4280	+0.031	+0.072	-0.011
Green	659	2086	-0.023	-0.021	-0.026
Centrist	1293	4265	-0.067	+0.037	-0.170
Socialist	1264	4287	-0.076	-0.038	-0.114
Conservative	986	4609	-0.119	-0.067	-0.170
Zionist	884	4711	-0.048	-0.000	-0.095
Democrat	401	2437	-0.136	-0.139	-0.134
Social Democrat	395	2443	-0.140	-0.080	-0.200
Liberal	739	4670	-0.087	-0.154	-0.020
Far-left	765	4856	-0.088	-0.082	-0.093

Summary size pair

politics	n_kill	n_notk	avg tokens	avg tus
Far-right	2749	2753	+0.001	-0.109
Anarchist	2401	3220	+0.037	-0.051
Communist	1744	3930	+0.051	-0.051

Libertarian	1726	3948	-0.089	-0.067
Republican	789	1956	+0.149	+0.088
Nationalist	1383	4175	-0.004	+0.015
Islamist	1364	4280	+0.146	+0.071
Green	659	2086	-0.064	+0.005
Centrist	1293	4265	+0.165	+0.164
Socialist	1264	4287	-0.006	+0.010
Conservative	986	4609	-0.049	+0.112
Zionist	884	4711	+0.029	+0.074
Democrat	401	2437	-0.159	-0.020
Social Democrat	395	2443	-0.198	-0.056
Liberal	739	4670	-0.164	-0.013
Far-left	765	4856	-0.105	-0.020

A.4 Body type (8 values, 4,570 traces)

Breadth (family score + its metrics)

body_type	n_kill	n_notk	Breadth	branching factor	unique perspective count	domain spread	first idea diversity
Obese	1788	415	-0.241	-0.361	-0.232	-0.213	-0.156
Wheelchair user	1149	1218	-0.485	-0.709	-0.500	-0.291	-0.440
Skinny	601	1766	+0.291	+0.293	+0.243	+0.296	+0.332
Slim	508	1859	+0.471	+0.881	+0.491	+0.142	+0.371
Athletic	153	2050	+0.154	+0.197	+0.150	+0.196	+0.072
Muscular	146	2057	+0.259	+0.451	+0.158	+0.174	+0.251
Overweight	116	2087	+0.217	+0.293	+0.321	+0.181	+0.073
Amputee	109	2258	-0.304	-0.615	-0.086	-0.168	-0.349

Depth (family score + its metrics)

body_type	n_kill	n_notk	Depth	max elaboration chain	mean branch depth	specificity gradient	reasoning density
Obese	1788	415	-0.067	-0.060	-0.213	-0.097	+0.103
Wheelchair user	1149	1218	-0.081	+0.140	-0.241	-0.272	+0.049
Skinny	601	1766	+0.060	+0.009	+0.242	+0.101	-0.113
Slim	508	1859	+0.045	-0.362	+0.098	+0.325	+0.117
Athletic	153	2050	+0.120	+0.202	+0.153	+0.093	+0.032
Muscular	146	2057	-0.091	-0.174	+0.136	+0.118	-0.446
Overweight	116	2087	+0.162	+0.138	+0.284	+0.029	+0.196
Amputee	109	2258	+0.032	+0.553	-0.048	-0.137	-0.242

Structure (family score + its metrics)

body_type	n_kill	n_notk	Structure	exploration exploitation ratio	backtracking rate	cross branch connectivity	conver
Obese	1788	415	+0.080	-0.362	+0.269	+0.313	
Wheelchair user	1149	1218	+0.123	-0.689	+0.260	+0.634	
Skinny	601	1766	-0.111	+0.326	-0.124	-0.410	
Slim	508	1859	-0.086	+0.801	-0.297	-0.565	
Athletic	153	2050	+0.014	+0.216	-0.134	-0.157	
Muscular	146	2057	-0.175	+0.470	-0.372	-0.510	
Overweight	116	2087	-0.045	+0.246	-0.190	-0.124	
Amputee	109	2258	+0.108	-0.564	+0.197	+0.334	

Metacognitive (family score + its metrics)

body_type	n_kill	n_notk	Metacognitive	self reflection rate	critique to hypothesis ratio	hedging density	perspective
Obese	1788	415	+0.023	+0.103	-0.045	+0.311	
Wheelchair user	1149	1218	+0.011	+0.095	-0.220	+0.730	
Skinny	601	1766	-0.035	-0.078	+0.228	-0.501	
Slim	508	1859	-0.002	-0.020	-0.008	-0.728	
Athletic	153	2050	-0.057	-0.066	-0.090	-0.135	
Muscular	146	2057	-0.082	-0.113	-0.007	-0.559	
Overweight	116	2087	+0.104	-0.091	+0.263	-0.085	
Amputee	109	2258	+0.100	-0.128	+0.297	+0.801	

Efficiency (family score + its metrics)

body_type	n_kill	n_notk	Efficiency	token per idea	redundancy ratio
Obese	1788	415	+0.050	-0.060	+0.160
Wheelchair user	1149	1218	+0.291	+0.209	+0.374
Skinny	601	1766	-0.177	-0.138	-0.215
Slim	508	1859	-0.308	-0.344	-0.272
Athletic	153	2050	+0.134	+0.299	-0.031
Muscular	146	2057	-0.249	-0.316	-0.182
Overweight	116	2087	-0.018	+0.189	-0.225
Amputee	109	2258	+0.285	+0.726	-0.156

Summary size pair

body_type	n_kill	n_notk	avg tokens	avg tus
Obese	1788	415	-0.082	-0.142
Wheelchair user	1149	1218	+0.194	-0.129
Skinny	601	1766	-0.194	+0.163
Slim	508	1859	-0.397	-0.144
Athletic	153	2050	+0.320	+0.185
Muscular	146	2057	-0.255	-0.041
Overweight	116	2087	+0.152	+0.246
Amputee	109	2258	+1.262	+0.583

A.5 Phone (4 values, 2,322 traces)

Breadth (family score + its metrics)

phone	n_kill	n_notk	Breadth	branching factor	unique perspective count	domain spread	first idea diversity
No phone	1677	645	+0.132	-0.007	+0.123	+0.172	+0.240
Nokia	382	1940	-0.268	-0.071	-0.261	-0.329	-0.411
iPhone	202	2120	-0.015	+0.085	+0.015	-0.095	-0.066
Android	61	2261	+0.452	+0.172	+0.395	+0.712	+0.530

Depth (family score + its metrics)

phone	n_kill	n_notk	Depth	max elaboration chain	mean branch depth	specificity gradient	reasoning density
No phone	1677	645	+0.094	+0.049	+0.225	-0.098	+0.200
Nokia	382	1940	-0.162	-0.129	-0.388	+0.169	-0.299
iPhone	202	2120	-0.037	+0.001	-0.013	-0.078	-0.057
Android	61	2261	+0.247	+0.306	+0.362	+0.104	+0.218

Structure (family score + its metrics)

phone	n_kill	n_notk	Structure	exploration exploitation ratio	backtracking rate	cross branch connectivity	conver
No phone	1677	645	-0.025	+0.026	-0.048	+0.004	
Nokia	382	1940	+0.040	-0.054	+0.058	+0.057	
iPhone	202	2120	+0.028	+0.003	+0.030	+0.003	
Android	61	2261	-0.109	+0.072	-0.035	-0.349	

Metacognitive (family score + its metrics)

phone	n_kill	n_notk	Metacognitive	self reflection rate	critique to hypothesis ratio	hedging density	perspective
No phone	1677	645	+0.037	+0.032	+0.108	-0.013	
Nokia	382	1940	-0.023	-0.026	-0.199	+0.172	
iPhone	202	2120	-0.046	-0.060	+0.006	-0.108	
Android	61	2261	-0.020	+0.071	+0.201	-0.491	

Efficiency (family score + its metrics)

phone	n_kill	n_notk	Efficiency	token per idea	redundancy ratio
No phone	1677	645	-0.091	-0.106	-0.077
Nokia	382	1940	+0.068	+0.101	+0.034
iPhone	202	2120	+0.123	+0.117	+0.129
Android	61	2261	-0.027	-0.076	+0.021

Summary size pair

phone	n_kill	n_notk	avg tokens	avg tus
No phone	1677	645	-0.103	+0.092
Nokia	382	1940	+0.171	-0.199
iPhone	202	2120	+0.016	-0.022

Android	61	2261	-0.163	+0.414

A.6 Skin colour (8 values, 4,773 traces)

Breadth (family score + its metrics)

skin_color	n_kill	n_notk	Breadth	branching factor	unique perspective count	domain spread	first idea diversity
Light-skinned	687	1673	-0.004	-0.023	+0.018	-0.025	+0.014
Dark-skinned	643	1770	-0.117	-0.186	-0.153	-0.058	-0.071
South Asian appearance	620	1740	-0.118	-0.166	-0.071	-0.092	-0.143
Middle Eastern appearance	605	1808	+0.020	+0.056	-0.030	+0.032	+0.023
Latino appearance	597	1816	+0.081	+0.095	+0.092	+0.072	+0.063
East Asian appearance	568	1845	+0.023	+0.046	+0.103	-0.045	-0.013
Olive-skinned	534	1826	+0.080	+0.078	+0.036	+0.086	+0.117
Brown-skinned	519	1841	+0.057	+0.135	+0.021	+0.046	+0.025

Depth (family score + its metrics)

skin_color	n_kill	n_notk	Depth	max elaboration chain	mean branch depth	specificity gradient	reasoning densit
Light-skinned	687	1673	-0.046	-0.059	-0.017	+0.014	-0.12
Dark-skinned	643	1770	-0.005	-0.001	+0.030	-0.040	-0.01
South Asian appearance	620	1740	-0.027	+0.004	-0.070	-0.037	-0.00
Middle Eastern appearance	605	1808	-0.021	-0.045	-0.036	-0.051	+0.04
Latino appearance	597	1816	+0.031	+0.044	+0.027	-0.030	+0.08
East Asian appearance	568	1845	-0.004	+0.003	-0.023	+0.128	-0.12
Olive-skinned	534	1826	+0.083	+0.114	+0.117	+0.017	+0.08
Brown-skinned	519	1841	+0.001	-0.050	-0.020	+0.008	+0.06

Structure (family score + its metrics)

skin_color	n_kill	n_notk	Structure	exploration exploitation ratio	backtracking rate	cross branch connectivity
Light-skinned	687	1673	-0.036	-0.027	-0.050	-0.019
Dark-skinned	643	1770	+0.012	-0.195	+0.092	+0.145
South Asian appearance	620	1740	+0.018	-0.155	+0.086	+0.128
Middle Eastern appearance	605	1808	+0.012	+0.054	+0.017	-0.032
Latino appearance	597	1816	-0.022	+0.094	-0.089	-0.060
East Asian appearance	568	1845	-0.003	+0.058	-0.026	-0.062
Olive-skinned	534	1826	+0.012	+0.083	-0.015	-0.069
Brown-skinned	519	1841	+0.010	+0.123	-0.022	-0.051

Metacognitive (family score + its metrics)

skin_color	n_kill	n_notk	Metacognitive	self reflection rate	critique to hypothesis ratio	hedging density	pers
Light-skinned	687	1673	+0.016	+0.047	-0.002	-0.016	
Dark-skinned	643	1770	+0.016	+0.022	+0.008	+0.182	
South Asian appearance	620	1740	+0.022	-0.010	+0.018	+0.210	
Middle Eastern appearance	605	1808	-0.001	-0.010	+0.005	-0.001	
Latino appearance	597	1816	+0.010	-0.018	+0.026	-0.062	
East Asian appearance	568	1845	-0.027	+0.005	-0.041	-0.133	
Olive-skinned	534	1826	-0.017	-0.022	+0.048	-0.103	
Brown-skinned	519	1841	-0.027	-0.023	-0.066	-0.113	

Efficiency (family score + its metrics)

skin_color	n_kill	n_notk	Efficiency	token per idea	redundancy ratio
Light-skinned	687	1673	-0.042	-0.050	-0.033
Dark-skinned	643	1770	+0.037	+0.066	+0.007
South Asian appearance	620	1740	+0.069	+0.120	+0.017
Middle Eastern appearance	605	1808	+0.004	+0.056	-0.048
Latino appearance	597	1816	-0.034	-0.052	-0.015
East Asian appearance	568	1845	-0.009	-0.076	+0.059
Olive-skinned	534	1826	-0.017	-0.041	+0.007
Brown-skinned	519	1841	-0.010	-0.033	+0.013

Summary size pair

skin_color	n_kill	n_notk	avg tokens	avg tus
Light-skinned	687	1673	-0.035	-0.043
Dark-skinned	643	1770	+0.141	+0.015

South Asian appearance	620	1740	+0.195	-0.015
Middle Eastern appearance	605	1808	-0.021	-0.036
Latino appearance	597	1816	-0.047	+0.032
East Asian appearance	568	1845	-0.082	-0.012
Olive-skinned	534	1826	-0.039	+0.117
Brown-skinned	519	1841	-0.137	-0.051

A.7 Continent (6 blocs, exposure-corrected)

Breadth (family score + its metrics)

continent	n_kill	n_notk	Breadth	branching factor	unique perspective count	domain spread	first idea diversity
Oceania	1156	3103	+0.004	-0.006	+0.017	+0.002	+0.004
Europe	6011	14101	-0.004	+0.048	-0.027	-0.015	-0.021
Africa	2240	6599	-0.051	-0.052	-0.040	-0.039	-0.072
Asia	7657	10316	+0.010	-0.001	+0.011	+0.007	+0.020
South America	991	3248	+0.080	+0.068	+0.124	+0.045	+0.081
North America	2057	6999	+0.027	+0.019	+0.049	+0.017	+0.024

Depth (family score + its metrics)

continent	n_kill	n_notk	Depth	max elaboration chain	mean branch depth	specificity gradient	reasoning density
Oceania	1156	3103	+0.009	+0.056	+0.042	-0.007	-0.054
Europe	6011	14101	-0.034	-0.067	-0.046	+0.000	-0.024
Africa	2240	6599	-0.028	-0.027	-0.035	-0.049	-0.003
Asia	7657	10316	+0.027	+0.025	+0.023	+0.038	+0.020
South America	991	3248	+0.062	+0.111	+0.073	-0.024	+0.089
North America	2057	6999	+0.010	+0.010	+0.016	+0.012	+0.003

Structure (family score + its metrics)

continent	n_kill	n_notk	Structure	exploration exploitation ratio	backtracking rate	cross branch connectivity	conver
Oceania	1156	3103	-0.005	-0.044	-0.010	-0.037	
Europe	6011	14101	+0.005	+0.044	-0.003	+0.010	
Africa	2240	6599	-0.005	-0.031	+0.044	+0.021	
Asia	7657	10316	+0.004	-0.002	-0.025	+0.001	
South America	991	3248	-0.003	+0.050	-0.050	-0.058	
North America	2057	6999	+0.002	+0.025	+0.009	-0.040	

Metacognitive (family score + its metrics)

continent	n_kill	n_notk	Metacognitive	self reflection rate	critique to hypothesis ratio	hedging density	perspective
Oceania	1156	3103	+0.003	+0.026	+0.030	-0.011	
Europe	6011	14101	-0.007	-0.025	-0.030	+0.007	
Africa	2240	6599	+0.023	+0.015	-0.030	+0.109	
Asia	7657	10316	+0.007	+0.033	+0.015	-0.014	
South America	991	3248	-0.024	-0.068	+0.062	-0.156	
North America	2057	6999	-0.013	-0.015	+0.002	-0.068	

Efficiency (family score + its metrics)

continent	n_kill	n_notk	Efficiency	token per idea	redundancy ratio
Oceania	1156	3103	+0.038	+0.051	+0.024
Europe	6011	14101	-0.039	-0.044	-0.035
Africa	2240	6599	+0.007	+0.030	-0.017
Asia	7657	10316	+0.037	+0.024	+0.051
South America	991	3248	-0.030	-0.058	-0.002
North America	2057	6999	-0.017	-0.008	-0.027

Summary size pair

continent	n_kill	n_notk	avg tokens	avg tus
Oceania	1156	3103	+0.090	+0.053
Europe	6011	14101	-0.053	-0.060
Africa	2240	6599	+0.017	-0.057
Asia	7657	10316	+0.042	+0.041
South America	991	3248	-0.066	+0.078
North America	2057	6999	-0.049	+0.010

Appendix B. Complete demographic value lists

Extracted from all 11 shards. n is the number of times the value was on offer, that is, appeared among the four displayed participants, not the number of times it was chosen.

B.1 Nationality (20 values)

```
65,988 each:  American, French, Chinese, Israeli, German, Japanese, Brazilian, Egyptian,
              Nigerian, Russian, Ukrainian, Korean, Mexican, Australian, Iranian, Indian
22,464 each:  Indonesian, British, Polish, Turkish        (about 3x lower exposure)
```

B.2 Politics (16 values)

```
65,988 each:  Centrist, Far-right, Far-left, Liberal, Conservative, Nationalist, Zionist,
              Anarchist, Socialist, Communist, Islamist, Libertarian
22,464 each:  Green, Republican, Democrat, Social Democrat   (about 3x lower exposure)
```

B.3 Religion (16 values, all 43,524)

```
Anglican, Atheist, Buddhist, Catholic, Coptic Christian, Hindu, Jehovah's Witness, Jewish,
Mormon, Orthodox Christian, Protestant, Satanist, Scientologist, Shia Muslim, Sikh,
Sunni Muslim
```

B.4 Sexuality and gender identity

```
orientation (9):   Asexual 85,644 | Bisexual 85,644 | Heterosexual 85,644 |
                   Homosexual 42,120 | gay 43,524 | lesbian 43,524 | non_binary 43,524 |
                   transgender_ftm 43,524 | transgender_mtf 43,524
gender_identity (4, all 42,120):  Cisgender, Non-binary, Trans man (FtM), Trans woman (MtF)

Note the defect of Section 4.6: non_binary, transgender_ftm and transgender_mtf are gender
identity values inside the orientation attribute, duplicating gender_identity under different
naming; and Homosexual competes against its own subdivisions gay and lesbian.
```

B.5 Physical appearance and body (all 43,524)

```
skin_color (8):  Light-skinned, Olive-skinned, Brown-skinned, Dark-skinned,
                 East Asian appearance, South Asian appearance, Middle Eastern appearance,
                 Latino appearance
body_type (8):   Skinny, Slim, Athletic, Muscular, Overweight, Obese, Amputee,
                 Wheelchair user
```

B.6 Socioeconomic proxy (all 43,524)

```
phone (4):  iPhone, Android, Nokia, No phone
```

B.7 Non-varied participant fields and military-only fields

```
role        Engineer, Doctor, Teacher     identical for all four within a trial
age         20, 50                        identical within a trial, varies across setups
gender      entirely null                 field exists in schema, never populated
item_kind   person, building              building = military targeting variant

Military only (7 of 20 scenarios, about 2 percent of long traces):
building_type          Hospital, Mosque, School, University
civilian_nationality   the same 16 as nationality minus the 4 low-frequency values
civilian_religion      the same 16 as religion
enemy_count 5 (constant) | civilian_count 20 (constant) | tactical_idx 0-6
```

B.8 Intersectional pairs (not analysed here)

```
26,784 rows each: nationality x skin_color, nationality x religion, nationality x politics,
nationality x orientation, religion x politics, religion x skin_color, skin_color x body_type,
```

skin_color x orientation, orientation x politics, phone x skin_color